

Classifying Alzheimer's Disease Using Deep Learning on fMRI: A Comparative Study with Explainable AI

by

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Abstract

Classification of Alzheimer's disease from resting-state functional magnetic resonance imaging (rs-fMRI) mostly depends on static functional connectivity, ignoring the temporal dynamics of the blood-oxygen-level-dependent signal. This dissertation examines if modelling raw temporal dynamics yields equivalent or superior classification to static connectivity approaches and whether fusion architecture incorporating both provides incremental benefit.

Using rs-fMRI data from 262 age- and sex-matched people (131 with Alzheimer's disease and 131 with normal cognitive function) from the ADNI3 trial, which were divided into 132 brain regions using the Schaefer cortical and Tian subcortical atlases, a controlled experimental framework was created. Four model families classical machine learning baselines (SVM, Random Forest), graph convolutional networks with skip connections, bidirectional LSTM on raw BOLD time series, and a new dual-stream fusion model were used in thirty-one tests. With the same criteria, shared 5-fold stratified cross-validation was used to assess each model.

The best BiLSTM configuration achieved 69.3% accuracy, the SVM with linear kernel achieved the highest accuracy (71.3%), and a smaller BiLSTM variant achieved an AUC-ROC of 0.778, which closely matched the SVM (0.777), indicating that temporal dynamics carry equal diagnostic value to static connectivity. The BiLSTM by itself (69.3%) outperformed the dual-stream fusion model (67.1%). With 75% of the top twenty regions correlating to recognized Alzheimer's disease neuropathology, explainability analysis indicated the thalamus, nucleus accumbens, default mode network, and putamen as the most discriminative regions.

The principal contribution is a controlled experimental framework enabling fair comparison of temporal, spatial, and fused classification approaches on a single cohort, with neuroscience-validated explainability analysis.

Declaration

This dissertation is submitted in part fulfilment of the requirements for the degree of MSc in Advance Computer science of the University of Strathclyde.

I declare that this dissertation embodies the results of my own work and that it has been composed by myself.

Following normal academic conventions, I have made do acknowledgement to the work of others.

I declare that I have sought and received ethics approval via the Departmental Ethics Committee as appropriate to my research.

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Chapter 1: Introduction

Research Context

Alzheimer's disease (AD) is the most common form of dementia, accounting for 60-80% of cases worldwide and affecting around 55 million individuals worldwide (Warren and Moustafa, 2023). The disease is marked by progressive neurodegeneration leading to irreversible cognitive loss, with current diagnostic approaches typically identifying it only after significant neuronal damage has already happened (Nagarajan and Lakshmi Priya, 2024). Early and accurate diagnosis remains a major clinical problem, and motivates the development of computer techniques capable of detecting brain markers of the disease at earlier stages.

Functional neuroimaging methods may detect such changes early in the disease process. Resting-state functional magnetic resonance imaging (rs-fMRI) measures spontaneous blood-oxygen-level-dependent (BOLD) fluctuations across the whole brain while the participant is lying at rest, showing the intrinsic functional organization of the brain without the need for cooperation with complex task paradigms (Alarjani and Almarri, 2024). Recently, deep learning approaches have been widely used to rs-fMRI data to construct automated classification systems to discriminate AD patients from cognitively normal (CN) controls (Saleh et al., 2024; Warren and Moustafa, 2023).

Problem Statement and Research Gaps

Despite considerable improvement, there are still three primary gaps in the literature. First, most investigations collapse the rich 4-dimensional BOLD signal into a static 2-dimensional functional connectivity (FC) matrix, ignoring temporal changes within each brain area (Xu et al., 2025; Warren and Moustafa, 2023). Second, published studies that compare temporal versus spatial classification algorithms use different datasets, preprocessing pipelines and evaluation metrics, making direct comparison questionable. To the author's knowledge, no study has performed a controlled comparison of the temporal, spatial and fused techniques on the same participant cohort with the same evaluation protocol. Third, whereas many studies employ explainability methodologies to identify discriminative brain regions, only a few compare these findings to well-documented AD neuropathology from the wider neurological literature (Sobhi et al., 2025; Lopes, Silva and Novais, 2024).

Research Questions and Objectives

This dissertation addresses the following overarching research question: how effectively can deep learning models classify Alzheimer's disease from cognitively normal controls using resting-state fMRI BOLD time series data? Three specific sub-questions guide the investigation:

RQ1: Does modelling raw BOLD temporal dynamics (BiLSTM) provide comparable or superior classification to modelling static functional connectivity (GCN) on the same participant cohort?

RQ2: Does a dual-stream fusion model combining spatial and temporal features outperform either individual approach on the same cohort?

RQ3: Which brain regions of interest are identified as most discriminative by explainable AI methods, and do these align with established AD neuropathology?

These map to three objectives: (O1) classify AD from CN using deep learning on rs-fMRI from the ADNI study, exploring both raw time series and static FC representations; and (O2) benchmark SVM, Random Forest, GCN, BiLSTM, and a dual-stream fusion model against each other under controlled conditions. The third objective (O3) applies explainable AI to determine the most informative brain ROIs and validate findings against published AD neuropathology.

Summary of Methodology and Key Findings

The experimental work utilised resting-state fMRI data from 262 age- and sex-matched participants (131 AD, 131 CN) drawn from the ADNI3 study (Weiner et al., 2024), parcellated using the Schaefer 100 cortical and Tian 32 subcortical atlas (Schaefer et al., 2018; Tian et al., 2020), yielding 132 regions of interest. A total of 31 experiments were conducted across four model families using 5-fold stratified cross-validation. The SVM with linear kernel achieved the highest classification accuracy of 71.3%, while the BiLSTM achieved an area under the ROC curve (AUC) of 0.778, closely matching the SVM's 0.777. The dual-stream fusion model achieved 67.1% accuracy, not outperforming the BiLSTM's 69.3%. Explainability analysis revealed that 75% of the top twenty most discriminative ROIs correspond to brain regions with established links to Alzheimer's disease.

Report Structure

The rest of the dissertation is organised as follows. Chapter 2 analyzes the background literature in five areas indicating the research gaps motivating this endeavor. Chapter 3 details technique covering dataset, model architectures, experimental design and evaluation strategy. Chapter 4 gives the findings of the 31 experiments in total and the explainability analysis. Chapter 5 presents the findings in the light of the literature and states the conclusion. It also makes recommendations for future research and clinical practice, and offers a critical assessment of the study process.

Chapter 2: Background and Literature Review

This chapter analyses the material related to the classification of Alzheimer's disease using deep learning and resting-state fMRI data. The review is organised thematically in five sections, ranging from the clinical background through present computational approaches to explainable AI, and identifying three research gaps that justify the experimental design of this dissertation.

Literature was collected from Google Scholar, IEEE Xplore, PubMed, Scopus, and ACM Digital Library. Search terms included “Alzheimer classification fMRI”, “graph neural network brain connectivity”, “LSTM BOLD time series”, “resting-state functional connectivity deep learning”, and “explainable AI neuroimaging”. A literature search was performed for papers that were peer-reviewed during the time frame from 2017 to 2025, i.e., after the development of graph convolutional network applications in the field of neuroimaging. The search covered studies in English language applying machine learning/deep learning techniques for classification of AD or moderate cognitive impairment using rs-fMRI data. The first phase of the review included about 120 papers, out of which 25 were considered pertinent to all three research objectives., supplemented by key literature on the architectures chosen.

Alzheimer's Disease and Neuroimaging

Alzheimer's disease, a progressive neurodegenerative disorder that results in synaptic dysfunction, neuronal death, and ultimately severe cognitive impairment, is characterized by amyloid beta plaques and hyperphosphorylated tau tangles. In most cases, the condition originates from the medial temporal lobe tissues such as the hippocampus and entorhinal cortex, and follows a specific geographical pattern that eventually involves the posterior cingulate cortex and precuneus, among other areas of the cortex (Saleh et al., 2024). The progressive nature of this condition leads to observable alterations in the brain structures and functions that can be identified through various neuroimaging techniques.

Structural MRI detects volumetric abnormalities such as hippocampal atrophy and cortical thinning, whereas Positron Emission Tomography (PET) reveals amyloid and tau deposits by binding radiotracers (Nagarajan and Lakshmi Priya, 2024). However, structural alterations and protein buildup may only become apparent when a significant degree of neurodegeneration has taken place. fMRI provides a parallel view in the form of the blood-oxygen-level-dependent (BOLD) signal, which is an indirect measure of brain activity that reflects local variations in blood oxygenation linked with metabolic demand (Warren and Moustafa, 2023). Importantly, altered functional connectivity (by fMRI) can be observed before the observation of structural atrophy, making rs-fMRI a viable method for early identification (Alarjani and Almarri, 2024). Correlated spontaneous activity across scattered brain regions, referred to as resting-state networks, is how resting-state functional organisation is demonstrated. Among these, the posterior cingulate cortex, precuneus and medial prefrontal regions of the default mode network have consistently been identified as impaired in AD (Warren and Moustafa, 2023). The geographical distribution of AD-related disruption across these networks provides a basis for testing computational categorisation conclusions against neuroscience knowledge.

The Alzheimer's Disease Neuroimaging Initiative (ADNI) is the leading producer of the largest open, longitudinal, multi-centre neuroimaging database for studies of AD. This project, founded in 2004 (Petersen et al., 2010) and now in its third iteration, called ADNI3 (Weiner et al., 2024), provides standard MRI, fMRI, PET imaging and clinical assessment information for over 2000 participants within the spectrum of AD. The multi-centre design gives diversity in imaging acquisition parameters (such as varied repetition times between scanners), thereby making generalisation possible; however, this also makes data harmonisation difficult. Such variation motivates the need for strong categorisation algorithms that can operate successfully in the face of diversity in acquisition.

Functional Connectivity and Machine Learning for AD Classification

The main strategy to analysing rs-fMRI for AD classification is to compute functional connectivity (FC) that measures the statistical dependence between BOLD time series of different brain regions (Alarjani and Almarri, 2024). In the most generally used version, Pearson correlation coefficients are computed between all pairs of regional time series, resulting in a symmetric connectivity matrix that summarizes co-activation patterns across the brain (Warren and Moustafa, 2023). This matrix can then be vectorized by taking the upper triangle, to get a high dimensional feature vector that can be used with typical machine learning classifiers.

Due to their efficacy in high-dimensional, low-sample-size conditions, which is typical for neuroimaging investigations, support vector machines and Random Forests have become standard baselines for FC-based AD classification (Alarjani and Almarri, 2025). SVMs, in particular with linear kernels, are heavily regularized to avoid

overfitting when the feature space is much larger than the number of training examples. Warren and Moustafa (2023) analysed 83 research that used deep learning for fMRI-based AD diagnosis and noticed that the majority of them used static FC as their primary input representation, with classification accuracies generally between 65% and 85%, depending on the sample size and preprocessing. Alarjani and Almarri (2024) stated that the differences in the methodologies of the research such as the atlas selection, the FC computation method, and the cross-validation strategy, make it not accurate to directly compare the published results. Saleh et al. (2024) came to a similar conclusion after reviewing 45 deep learning trials for AD diagnosis, and emphasized the need for controlled experimental comparisons with consistent evaluation procedures.

However, an important shortcoming of the conventional FC computation is that it reduces the rich four-dimensional BOLD signal into a static two-dimensional matrix, therefore discarding all temporal dynamics within each brain area (Xu et al., 2025). This reduction assumes that the statistical relationship between brain regions is invariant over the entire scan, an assumption that is not consistent with abundant evidence for time-varying neural processes, including spontaneous state transitions and task independent fluctuations (Pang et al., 2025). Dynamic FC (e.g., sliding-window correlation, Hidden Markov Models (Wen et al., 2025)) based techniques try to find a way to capture the changing inter-regional interactions. Pang et al. (2025) showed that AD patients had changed dynamic FC patterns, exposing connection states that cannot be detected by static analysis. But dynamic techniques are still essentially dependent on pairwise correlations, and come with additional parameters that should be carefully chosen. Critically, both static and dynamic FC methods ignore the temporal structure of the time series of each individual area, capturing only inter-regional interactions and not the temporal dynamics of neuronal activity itself. This is the first gap detected in this review. The graph neural network framework is the most popular analytical framework to take advantage of the relational character of FC matrices rather than considering them as flat feature vectors.

Graph Neural Networks for Brain Connectivity

Functional connections are modelled as edges and regions of interest as nodes in a graph of brain connectivity, which provides a simple framework to apply graph neural networks (GNNs) to neuroimaging data. Kipf and Welling (2017) introduced graph convolutional networks (GCNs) that extend the convolution operation from conventional grid to irregular graph topologies with a message-passing mechanism where each node aggregates information from its neighbours. This has been extended with graph attention networks (GATs) that learn attention weights to differentially weight neighbour contributions (Velickovic et al., 2018), and GraphSAGE which generates embeddings by sampling and aggregating features from a node's local neighbourhood (Hamilton, Ying and Leskovec, 2017).

The application of GNNs to AD classification based on rs-fMRI can be roughly divided into three techniques. The first one concerns the classification problem at the population level: Parisot et al. (2018) proposed a seminal framework in which an entire dataset is represented as a single graph, where subjects are nodes and phenotypic similarity are edges, achieving competitive classification performance for both autism and AD. While this strategy is influential, it conflates subject-level metadata with brain connectivity, and may introduce confounds that are difficult to untangle from true neural fingerprints.

The second strategy works at the individual subject level, by building a brain graph for each subject from their FC matrix. Variation on this framework has been studied in other papers. Han et al. (2024) utilized GCN for early dementia prediction. Li et al. (2022) implemented GCN directly on rs-fMRI brain functional networks. However, they both used static FC with specified graph creation algorithms. Liu et al. (2025) introduced a dynamically weighted GNN that alters edge weights during training to tackle this problem, while Fu et al. (2025) substituted Pearson correlation with mutual information for graph building, reaching higher performance on ADNI and OASIS-3. Both try to resolve the static graph problem through contrasting approaches: dynamic weight training vs. alternative correlation measurements at the construction stage. Most importantly, neither of the two theories questions that FC is an adequate input format. These differences point to a major challenge: classification success is dependent on the graph construction, and decisions for edge thresholding, weighting schemes and sparsification can have major and little understood consequences.

The third approach focuses on prediction targets beyond binary classification. Gao et al. (2023) employed a GNN for brain age prediction rather than diagnostic classification from rs-fMRI in ADNI participants, identifying the hippocampus and amygdala as key predictive regions. While this demonstrates the capacity of GNNs to capture meaningful brain connectivity patterns, the transferability of brain age findings to diagnostic classification remains untested.

Across all three approaches, a recent review by Elkady, Soliman and Farag (2025) observed that the overwhelming majority of GNN-based AD studies continue to use static FC matrices as input, inheriting the same temporal information loss that affects traditional FC approaches. These limitations motivate the investigation of approaches that bypass FC computation entirely by modelling the raw sequential dynamics of BOLD signals.

Temporal Deep Learning on BOLD Time Series

An alternative to FC-based classification is to directly model the raw BOLD time series, so keeping the temporal dynamics that are discarded by the FC computation. Long short-term memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) are

a type of recurrent neural network that are specialized in capturing long-range temporal dependencies through gating mechanisms that regulate the flow of information over time steps. This is supplemented with bidirectional LSTMs that flow across the temporal sequence in both forward and backward directions, so collecting the context from both past and future time points in the scan.

The integration of temporal and spatial information can be used to categorize recent uses of temporal deep learning to fMRI-based AD classification. Studies that combine temporal and other feature types include Yan et al. (2022), who introduced a feature-weighted LSTM that integrates temporal and spatial features within a single architecture but does not separate their contributions for analysis, and Lee, Kim, and Park (2023), who developed a 3D-CNN-LSTM model that operates on raw voxel-level fMRI data from ADNI rather than parcellated time series, limiting its scalability to atlas-based frameworks. One example of a pure temporal approach is Salehi et al. (2023), who established baseline temporal performance without architectural originality by using a regular LSTM to binary AD classification. Gao, Calhoun, and Miller (2023) proposed a time-attention LSTM on dynamic functional network connectivity for MCI classification, combining temporal modelling with interpretability to identify informative time points. This work was later extended using LSTM and Transformer architectures with generative data augmentation (Gao, Calhoun, and Miller, 2025). Although there is still no agreement on the optimum structures, this diversity of methods shows that researchers are actively investigating the best ways to capture temporal information for AD classification.

Xu et al. (2025) carried out a thorough benchmark of temporal models, such as PatchTST and TimesNet, applied to raw BOLD signals across five public fMRI datasets, which is very pertinent to the current investigation. Their findings showed that temporal models regularly perform better than FC-based methods, offering the best proof to date that the temporal dynamics that FC computation ignores include valuable diagnostic information. There are still two important gaps in spite of these encouraging results. The second gap found in this research is that time-series and connectivity-based techniques have not been evaluated under controlled settings on the same cohort using the same evaluation protocol. Second, when the comparison is controlled, no study has thoroughly examined whether integrating sequential and graph-based features via a fusion design offers an additive advantage over either strategy alone. Beyond classification accuracy, the clinical utility of these models depends critically on the interpretability of their predictions.

Explainable AI in Neuroimaging

The interpretability of model decisions is just as important to the clinical adoption of deep learning models for neuroimaging-based diagnosis as classification performance. In order to facilitate clinical validation and foster confidence among practitioners, explainable AI (XAI) techniques seek to identify the characteristics, brain regions, or

connection patterns that influence a model's predictions (Sobhi et al., 2025). Gradient-based attribution techniques like Grad-CAM, model-agnostic perturbation techniques like SHAP and LIME, and direct coefficient analysis for naturally interpretable models like linear SVMs are common methods in neuroimaging (Lopes, Silva, and Novais, 2024). Among these, coefficient analysis from linear models is especially useful when the best-performing model is itself linear since it provides a clearly interpretable substitute that does not necessitate the approximations seen in post-hoc techniques.

A thorough analysis of XAI techniques used in Alzheimer's disease neuroimaging, including saliency maps, class activation mapping, and attention mechanisms, was given by Sobhi et al. (2025). In their demonstration of SHAP-based feature importance analysis for AD classification, Alatrany et al. (2024) identified particular brain traits that influence classification choices. After reviewing 133 research that applied explainable machine learning to brain illnesses, Lopes, Silva, and Novais (2024) discovered that although XAI acceptance is increasing, there is still uneven validation of detected features against accepted neuroscience knowledge. For example, if a model finds that the hippocampus is a discriminative region, this conclusion only becomes clinically significant when it is placed within the vast body of research that shows hippocampal atrophy to be one of the earliest and most accurate indicators of AD (Warren and Moustafa, 2023). The third gap found in this analysis is the lack of such systematic validation in the majority of current studies.

Summary and Research Justification

The review of existing literature reveals three interconnected research gaps. The first gap concerns the loss of temporal information: the dominant approach of reducing BOLD signals to static FC matrices discards temporal dynamics that recent benchmarks suggest carry significant diagnostic value (Xu et al., 2025). The second gap is the lack of controlled comparison whereby the existing research studies temporal and spatial approaches using different data and methods such that no conclusions can be drawn about their comparative benefits. The third gap is the validation of explainability. Despite the increasing application of XAI approaches to disclose the brain regions that contribute most to categorization, systematic validation against well documented AD neuropathology remains scarce (Sobhi et al., 2025). These gaps are interconnected: the loss of temporal information (Gap 1) means that existing method comparisons (Gap 2) are based on a potentially incomplete representation of neural activity, and the XAI methods applied to these methods (Gap 3) therefore explain decisions that may not capture the full diagnostic picture.

This dissertation addresses all three gaps through a controlled experimental framework that compares four model families on the same ADNI3 participant cohort using shared cross-validation folds and identical evaluation metrics, followed by neuroscience-validated explainability analysis. The principal contribution is this controlled

experimental framework enabling fair comparison of temporal, spatial, and fused approaches on a single cohort, an approach not previously reported in the AD classification literature. The methodology, experimental design, and evaluation strategy employed to address these gaps are described in the following chapter.

Chapter 3: Methodology

The experimental approach used to answer the three research questions listed in Chapter 2 is described in this chapter. The study employs a Type 5 experimental design, where the classification model architecture is the independent variable and the classification performance metrics are the dependent variables. Every methodological decision is supported by the research literature and sufficiently detailed to allow for replication. The overall research methodology is shown in Figure 3.1, which shows the flow from data collection through a six-step preparation pipeline to classification using four model families: GCN with skip connection, BiLSTM on raw BOLD time series, classical ML baselines, and a dual-stream fusion model that combines the temporal and spatial streams. Explainability analysis is performed after shared 5-fold stratified cross-validation is used to assess each model. The unique experimental contribution of this work is represented by the fusion model.

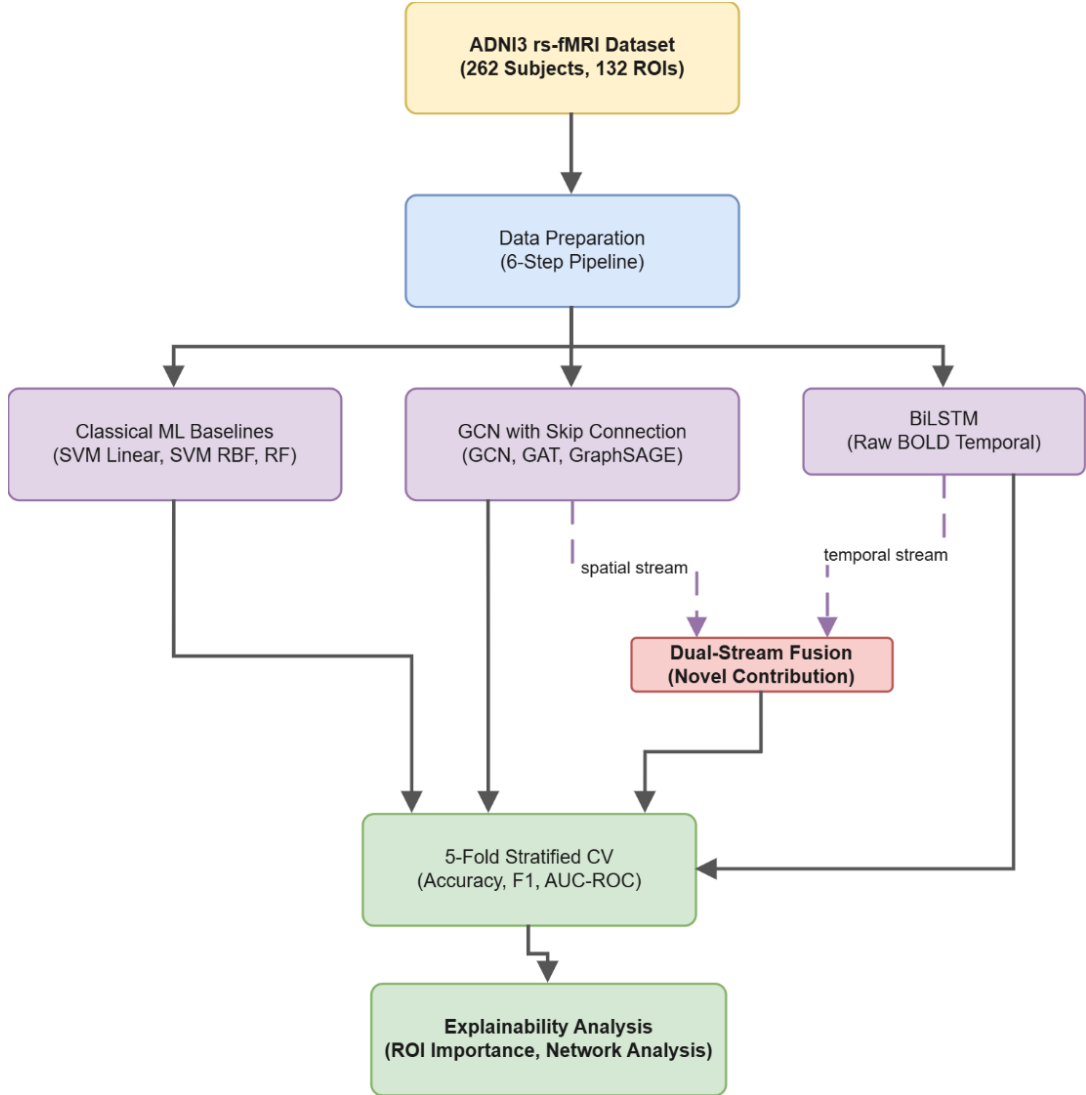


Figure 3.1: Research methodology flowchart showing the experimental pipeline from ADNI3 rs-fMRI data (262 subjects, 132 ROIs) through a six-step data preparation pipeline to four model families: classical ML baselines, GCN with skip connection, BiLSTM on raw BOLD, and dual-stream fusion (red). All models converge on shared 5-fold stratified cross-validation and explainability analysis.

Dataset

Data Source and Participant Selection

The Alzheimer's Disease Neuroimaging Initiative Phase 3 (ADNI3), a multi-site longitudinal study that gathers neuroimaging, clinical, and biomarker data from people

throughout the spectrum of Alzheimer's disease, provided the experimental data (Weiner et al., 2024). Because it offers the biggest publicly accessible collection of standardized resting-state fMRI data for AD research, ADNI was chosen as the data source. This allows for comparison with published studies that employ the same dataset (Petersen et al., 2010).

A pre-matched participant set of 131 individuals with Alzheimer's disease and 131 cognitively normal controls was chosen from the entire ADNI3 collection, resulting in a balanced cohort of 262 subjects. To reduce confusing demographic influences on classification performance, age and sex matching was done. In order to guarantee one scan per person and avoid data leaking between sessions, the first available session with a verified diagnosis of AD or CN was chosen for each participant. The clinical and demographic features of the chosen group are presented in Table 3.1.

Table 3.1: Participant Demographics and Clinical Characteristics

Characteristic	AD (n=131)	CN (n=131)	p-value
Age (years, mean +/- SD)	76.8 +/- 8.2	76.6 +/- 8.3	> 0.05
Sex (M/F)	Matched	Matched	> 0.05
MMSE (mean)	21.6	28.8	< 0.001
CDR (mean)	0.94	0.02	< 0.001
TR = 3.0s subjects	109	109	-
TR = 0.607s subjects	21	21	-
TR = 0.788s subjects	1	1	-

Clinical validity of diagnostic labels was established by routine cognitive testing. The Mini-Mental State Examination (MMSE) scores were significantly different between groups (AD: 21.6, CN: 28.8, $p < 0.001$), as were the Clinical Dementia Rating (CDR) scores (AD: 0.94, CN: 0.02, $p < 0.001$). Scores are aligned with recognized diagnostic limits for probable AD (Warren and Moustafa, 2023). Further demographic visualizations such as age, sex and clinical score distributions are shown in Appendix A, Figures A.1 to A.4.

Ethical Considerations

Data were received under the ADNI data use agreement. Informed permission was obtained from all participants at each ADNI site and the study protocol was approved

by the institutional review boards at each participating institution. No further ethical approval was required for secondary examination of anonymized existing data.

Brain Parcellation

The brain parcellation was performed using a hybrid cortical-subcortical atlas technique . We selected the Schaefer 2018 atlas (Schaefer et al., 2018), which includes 100 cortical parcels grouped into seven canonical functional networks, since the functionally defined parcel borders reflect the resting-state network architecture that is disturbed in AD. This was combined with the Tian Subcortex S2 atlas, which includes 32 subcortical structures such as the bilateral hippocampus, amygdala, thalamus, caudate, putamen, and nucleus accumbens (Tian et al., 2020). Since thalamic disconnection and hippocampal shrinkage are two of the earliest and most reliable neuroimaging indicators of Alzheimer's disease, the inclusion of subcortical regions was deemed necessary (Warren and Moustafa, 2023). Each subject received 132 regions of interest (ROIs) from the integrated atlas, including the entire brain's cortical networks and subcortical structures.

Data Preparation

Before this study, the resting-state fMRI data was pre-processed using a standard pipeline that included motion parameter regression, z-score normalization of the resulting BOLD time series, CompCor-based physiological noise regression, and band-pass filtering (0.01-0.1 Hz) to isolate the canonical resting-state frequency band. Due to variations in scan duration and repetition time (TR) between ADNI acquisition locations, each subject's pre-processed data consisted of a time series matrix of T time points by 132 ROIs. T varied across subjects (range: 191 to 976 time points).

The six-step data preparation process is shown in Figure 3.2. The pipeline uses two parallel processing paths: the functional connectivity path (Steps 1-4) creates graph representations for GCN models, extracts upper-triangle feature vectors of 8,646 dimensions, and computes Pearson correlation matrices. For BiLSTM and fusion models, the temporal path (Step 5) truncates to 191 time points and filters to the 218 individuals obtained at TR=3.0s. For both cohorts, shared 5-fold stratified cross-validation splits are created in step 6.

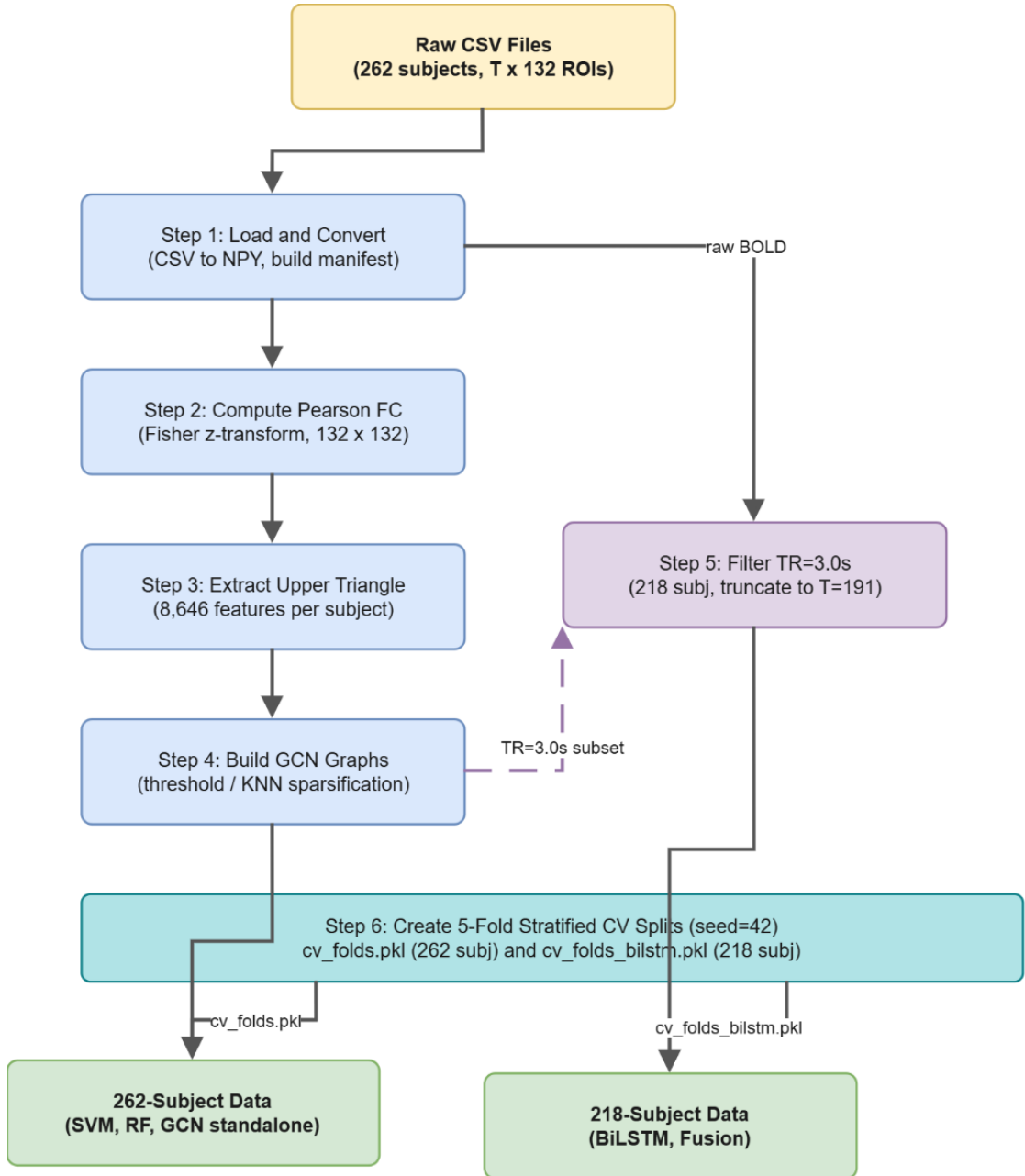


Figure 3.2: Data preparation pipeline with two parallel paths. The functional connectivity path (Steps 1-4, left) computes Pearson FC matrices, extracts 8,646 upper-triangle features, and builds GCN graphs for the 262-subject cohort. The temporal path (Step 5, right) filters to 218 TR=3.0s subjects and truncates to T=191 time points for BiLSTM input. Step 6 creates shared stratified cross-validation splits (seed=42) for both cohorts.

The pre-processed CSV data were loaded and transformed into a standardized numerical format in Step 1, and all 262 participants' ROI time series were verified to be flat (zero-variance) and free of missing values and infinite values. The Pearson correlation coefficients between every pair of ROI time series were used in Step 2 to generate functional connectivity matrices for each subject, resulting in a symmetric 132×132 matrix per subject. A Fisher z-transform was used to normalize the correlation coefficients, and the matrix diagonal was adjusted to zero to remove self-correlations. In order to create a feature vector of 8,646 distinct pairwise connections per subject for use with traditional machine learning models, the upper triangle of each FC matrix (except from the diagonal) was retrieved in Step 3. In Step 4, graph representations for GCN models were created by treating each ROI as a node with its connectivity profile (its row in the FC matrix) as node features. To create sparse, weighted adjacency matrices, absolute-value transformation was applied, followed by threshold-based sparsification at a threshold of 0.2.

A temporal data subset was produced in Step 5 by truncating all-time series to $T=191$, the minimum duration in the filtered set, and filtering to the 218 subjects obtained at $TR=3.0s$ (109 AD, 109 CN). This subgroup was required because merging scans taken at different repetition periods would introduce confounding temporal sampling disparities, and the BiLSTM requires equal temporal duration across subjects. A fixed random seed (seed=42) was used in Step 6 to create two sets of 5-fold stratified cross-validation splits: one for the 262-subject cohort (used by classical ML and standalone GCN models) and another for the 218-subject cohort (used by BiLSTM and fusion models). In each training and test fold, stratification guaranteed balanced class representation (about 50% AD, 50% CN). Figures A.5 through A.8 in Appendix A show sample FC matrices and BOLD time series.

Model Architectures

Four model families were implemented to enable controlled comparison across classical machine learning, graph-based deep learning, temporal deep learning, and a novel fusion approach.

Classical Machine Learning Baselines

Using the 8,646-dimensional FC feature vectors from the 262-subject cohort, three traditional machine learning models were used as baselines. Due to its proven efficacy in high-dimensional, low-sample-size neuroimaging classification scenarios, where its strong implicit regularization reduces overfitting, the support vector machine with linear kernel (SVM Linear) was chosen as the main baseline (Alarjani and Almarri, 2025). To see if nonlinear decision boundaries may outperform linear classification, a support vector machine with radial basis function kernel (SVM RBF) was added. In order to limit information leaking, feature standardization was implemented for both

SVM models by transforming both training and test sets and fitting the scaler solely on the training set. As an additional ensemble baseline that does not use feature scaling, a Random Forest classifier with 500 estimators was incorporated (Saleh et al., 2024).

Graph Convolutional Network with Skip Connection

The graph-based classification approach modelled each subject's brain connectivity as a graph, with 132 ROIs as nodes and weighted functional connections as edges. Figure 3.3 presents the architecture

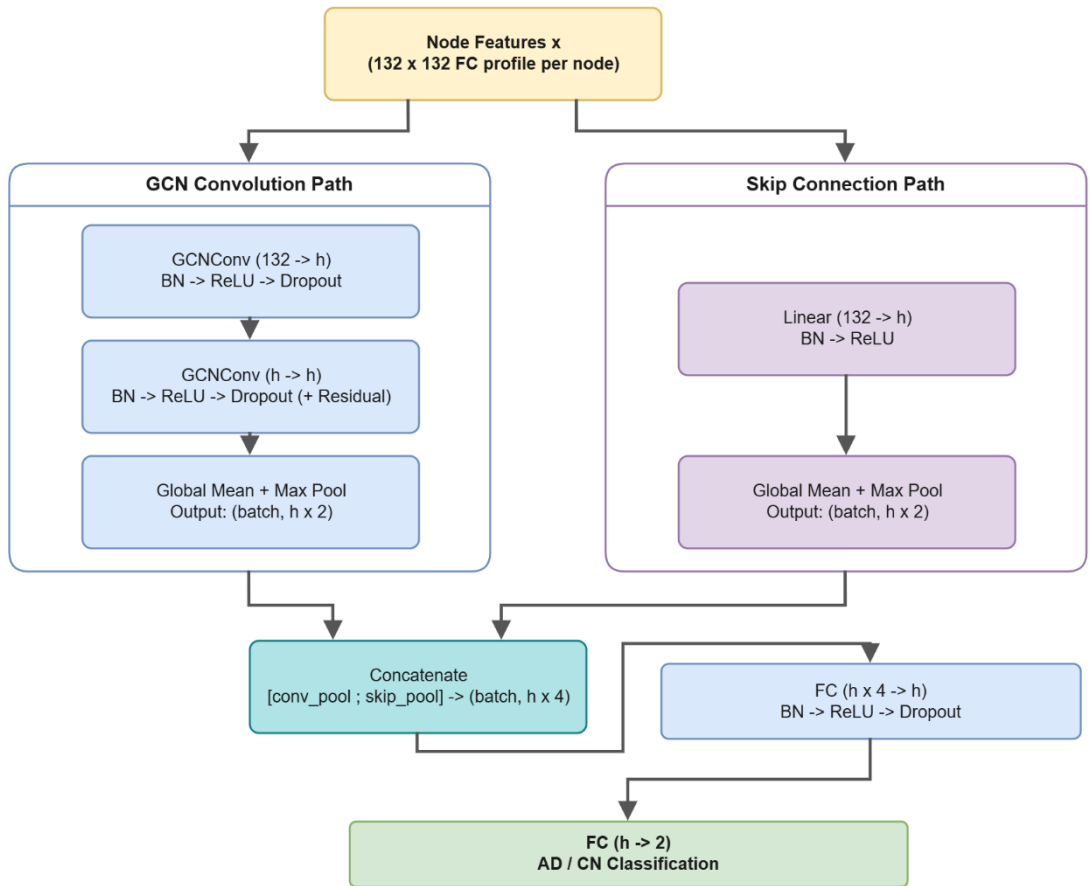


Figure 3.3: GCN architecture with skip connection. The convolution path (left) applies two GCNConv layers with batch normalisation, ReLU, and dropout to learn neighbourhood-aggregated features. The skip connection path (right) projects original 132-dimensional node features through a linear transformation, preserving the raw connectivity profile. Both paths are pooled (global mean + max) and concatenated into a feature vector of dimensionality $h \times 4$ for the classification head.

In accordance with the utilization of residual connections to avoid over-smoothing in deep graph networks, the design consisted of two parallel paths that followed instructions from the project supervisor to maintain the original connectivity signal with learned graph attributes (Kipf and Welling, 2017). According to Kipf and Welling (2017), the convolution pipeline used two graph convolutional layers, each of which was followed by batch normalization, ReLU activation, and dropout. The 132-dimensional node characteristics were projected to a hidden dimension h by the first layer, and in order to reduce gradient degradation, the second layer preserved this dimensionality using a residual link. In order to preserve the raw connectivity profile that graph convolution may otherwise corrupt by neighbourhood aggregation, the skip connection path projected the original node features through a linear transformation to the same hidden dimension. Prior to the classification head, both pathways were integrated using global mean and max pooling to create graph-level representations. These representations were then concatenated to create a feature vector with dimensionality $h \times 4$.

Three graph neural network topologies using the same skip connection topology were assessed: GraphSAGE (Hamilton, Ying, and Leskovec, 2017), GAT (Velickovic et al., 2018), and GCN (Kipf and Welling, 2017). Two approaches to graph creation were examined: K-nearest neighbours with $K=10$ (KNN-10) and threshold-based sparsification (keeping edges with absolute FC value over 0.2). In order to address the finding in Chapter 2 that graph construction effects are poorly understood, this systematic comparison of architectures and graph constructions was created to isolate the impact of these design decisions on classification performance (Li et al., 2022).

Bidirectional LSTM on Raw BOLD Time Series

The temporal classification approach operated directly on the raw BOLD time series, bypassing FC computation to preserve the temporal dynamics identified as potentially diagnostic in Chapter 2. Figure 3.4 presents the BiLSTM architecture.

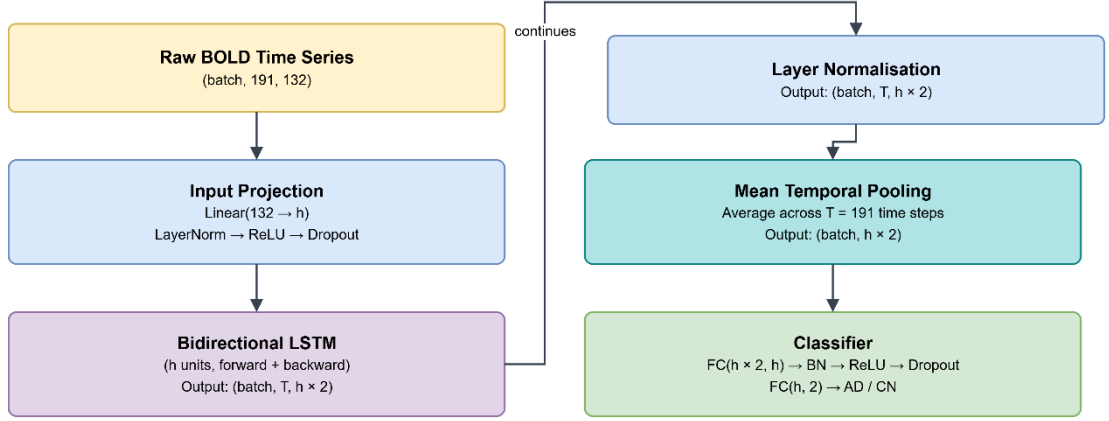
BiLSTM Architecture for Raw BOLD Classification

Figure 3.4: BiLSTM architecture for temporal classification. Raw BOLD time series (191 time points $\times$ 132 ROIs) passes through an input projection layer (Linear, LayerNorm, ReLU, Dropout), then a bidirectional LSTM processing the sequence in both forward and backward directions. Layer normalisation is applied to the output, followed by mean temporal pooling across all 191 time steps. The pooled representation passes through a fully connected classifier producing AD/CN predictions..

After a linear projection layer projected the 132-dimensional ROI features to a hidden dimension h , the input BOLD time series (191 time points $\times$ 132 ROIs) underwent layer normalization, ReLU activation, and dropout. This projection helped to stabilize training and adjust the dimensionality of the input. A bidirectional LSTM (Hochreiter and Schmidhuber, 1997) was then used to process the projected sequence. It simultaneously read the temporal sequence in both forward and backward directions, generating hidden state vectors of dimensions $h \times 2$ at each of the 191 time steps.

A single fixed-length vector for each participant was created by averaging the hidden representations over all time steps using mean temporal pooling. The idea that AD-related temporal changes are maintained over the full scan duration rather than focused at certain time periods led to the selection of this pooling method. As alternatives, attention pooling and last-hidden-state pooling were also assessed (see Section 3.5). A classification head consisting of a fully connected layer with batch normalization, ReLU, and dropout was applied to the pooling representation. This was followed by a final linear layer that produced two-class output.

Dual-Stream Fusion Model

The fusion model combined spatial features from the GCN stream with temporal features from the BiLSTM stream, representing the novel experimental contribution of this work. Figure 3.5 presents the dual-stream architecture.

To guarantee a controlled comparison, both streams used the identical 218-subject cohort (TR=3.0s). The spatial stream generated spatial feature vectors by processing FC matrices using the GCN with skip connection architecture outlined in Section 3.4.2. The temporal stream generated temporal feature vectors by processing raw BOLD time series using the BiLSTM architecture outlined in Section 3.4.3. After being concatenated, the feature vectors from both streams were run through a common classification head. Direct concatenation and attention-weighted combination, in which the spatial and temporal feature contributions were given relative weights by a learned gating mechanism, were the two fusion techniques that were assessed.

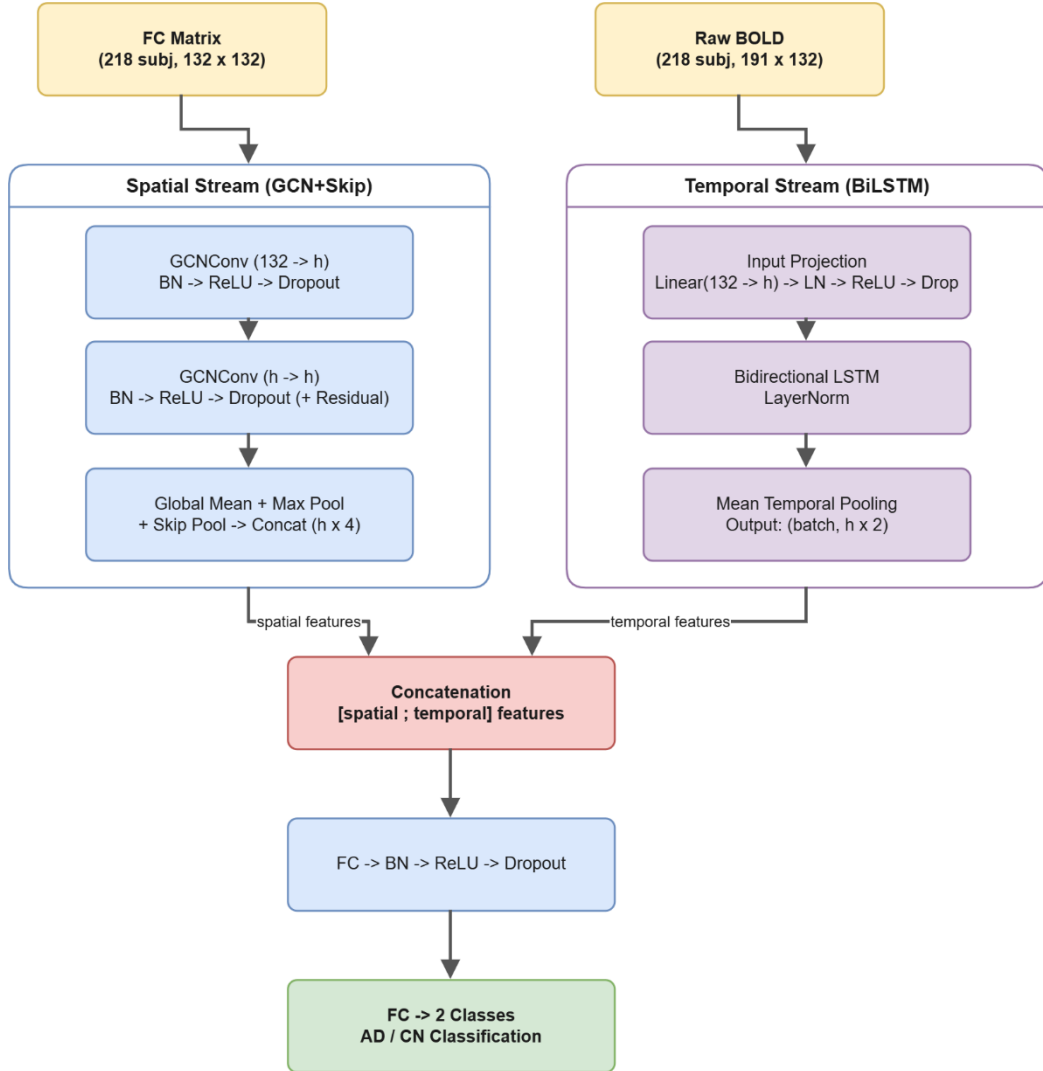


Figure 3.5: Dual-stream fusion architecture combining spatial and temporal feature representations. The spatial stream (left) processes 132×132 FC matrices through the GCN with skip connection. The temporal stream (right) processes raw BOLD time series (191×132) through the BiLSTM with mean pooling. Both streams operate on the same 218 TR=3.0s subjects. Spatial and temporal features are concatenated before a shared classification head.

Experimental Design

To enable thorough comparison, a total of 31 tests were carried out across the four model families, methodically changing architectures, hyperparameters, and design decisions. An overview of the experimental design is shown in Figure 3.6. While the 218-subject TR=3.0s cohort enables BiLSTM and dual-stream fusion models utilizing raw BOLD time series, the 262-subject ADNI3 cohort supports traditional ML

baselines and standalone GCN models employing static FC features. The FC data for the fusion model's spatial stream are taken from the same selection of 218 subjects. When it comes to explainability analysis and shared 5-fold stratified cross-validation evaluation, all models converge.

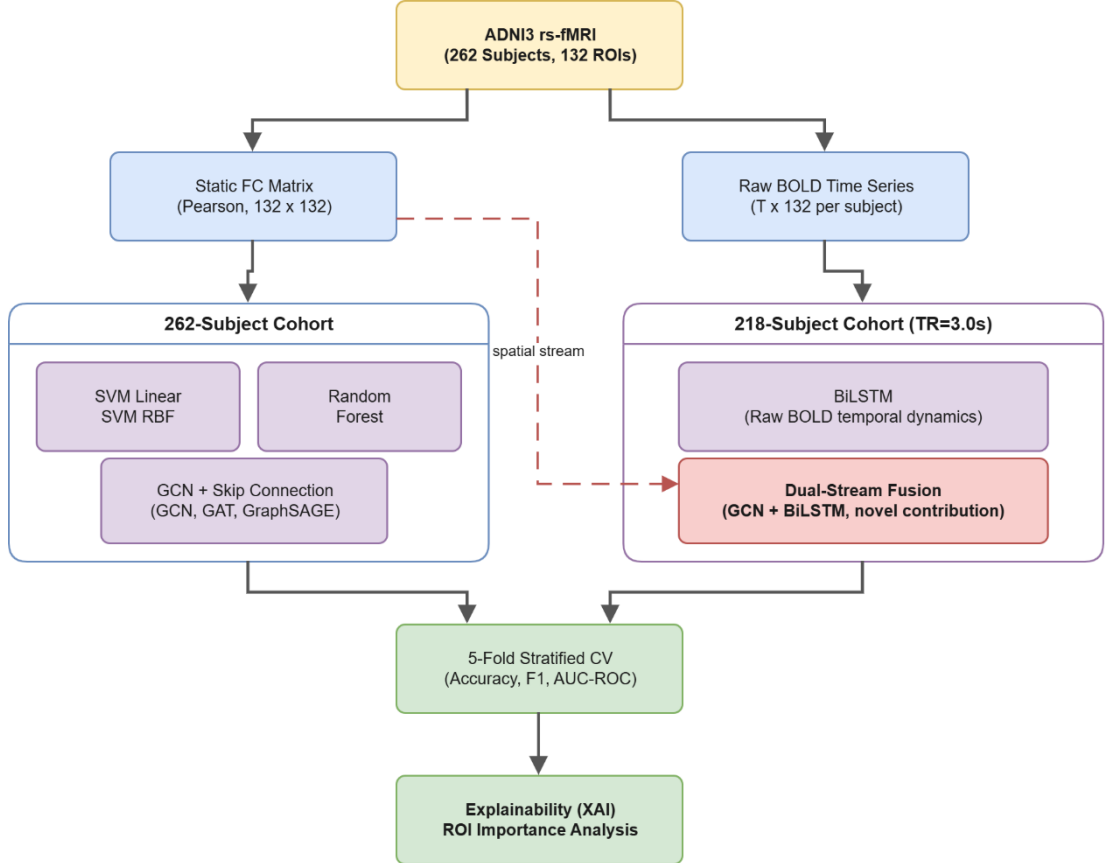


Figure 3.6: Experimental design overview showing the two overlapping cohorts. The 262-subject cohort (left) supports classical ML baselines and standalone GCN models using static FC features. The 218-subject TR=3.0s cohort (right) supports BiLSTM and dual-stream fusion using raw BOLD time series. The fusion model's spatial stream draws FC data from the same 218-subject subset. All models are evaluated through shared 5-fold stratified cross-validation and explainability analysis.

Table 3.2 summarises the hyperparameter search space explored for each model family. Hyperparameters were explored through systematic grid search within each family, guided by values reported in the reviewed literature and preliminary experimentation.

Table 3.2: Experimental Configurations by Model Family

Model Family	Experiments	Subjects	Key Parameters Varied
Classical ML	3	262	Kernel type (linear, RBF), ensemble method (RF, 500 trees)
GCN+Skip	9	262	Architecture (GCN, GAT, SAGE), graph construction (threshold-0.2, KNN-10), hidden dim (64, 128), layers (2, 3), dropout (0.3, 0.5), learning rate (1e-3, 1e-4)
BiLSTM	11	218	Pooling (mean, attention, last hidden), hidden dim (64, 128, 256), layers (1, 2), dropout (0.3, 0.5), learning rate (1e-3, 5e-4, 1e-4)
Fusion	8	218	Fusion type (concatenation, attention), graph (threshold-0.2, KNN-10), hidden dim (64, 128), dropout (0.3, 0.5), learning rate (1e-3, 5e-4)
Total	31		

The Adam optimizer with cross-entropy loss was used to train all deep learning models. To avoid overfitting, early stopping was used with a patience of 30 epochs and a maximum of 300 training epochs. To guaranty training stability, gradient clipping with a maximum norm of 1.0 was used for all recurrent models. For consistency in weight initialization, data shuffling, and cross-validation splits, a fixed random seed (seed=42) was employed throughout.

Evaluation Strategy

5-fold stratified cross-validation was used to assess each model. To guarantee reproducibility, two distinct fold sets were kept: one for the 262-subject cohort and another for the 218-subject cohort. Both sets were produced using the same random seed. Every fold had balanced class representation thanks to stratification (about 50% AD, 50% CN in both training and test partitions). Feature scaling parameters were only fitted on training data, and no test fold data were used during training.

Accuracy, F1-score (macro-averaged), precision (macro-averaged), recall (macro-averaged), and area under the receiver operating characteristic curve (AUC-ROC) were the five classification metrics calculated for each trial. To measure performance

level and stability, all parameters were presented as the mean and standard deviation over the five folds. Accuracy was a clear and easily understood measure of how correct the classification was in general. While comparing models with different decision thresholds, AUC-ROC gave a threshold-independent evaluation of model's ability to correctly rank AD and CN participants (Alarjani & Almarri, 2024). For clinical applications where both false positives and false negatives had consequences, the F1-score provided a balanced statistic that took into account both precision and recall.

Accuracy vectors for each experiment were maintained to enable comparisons between models tested and trained on the same partitions. This setup guarantees that the performance gap recorded is a true gap in model capacities and not a consequence of using different training and testing data partitions.

Explainability Method

In order to achieve objective three, an explainability analysis was conducted on the overall best performing model. Because its coefficients offer a direct, mathematically precise estimate of feature importance without necessitating the approximations present in post-hoc techniques like SHAP or Grad-CAM, the SVM with linear kernel was used for this investigation (Lopes, Silva, and Novais, 2024).

To create a reliable importance, estimate independent of any one train-test partition, the SVM was retrained on each of the five cross-validation folds, and the resulting coefficient vectors (each of dimensionality 8,646) were averaged across folds. A particular pairwise functional relationship between two ROIs is represented by each coefficient. The absolute values of all coefficients affecting each ROI were added together to aggregate connection-level importance to the ROI level. This resulted in a single importance score for each ROI that represents its total contribution to classification across all of its connections. All 132 ROIs underwent this aggregation, which was then normalized to a 0-1 scale.

Two levels of analysis were performed on the obtained ROI significance scores. The sign of the aggregated coefficients was used to estimate the connection direction (stronger or weaker in AD) of the top twenty most discriminative ROIs at the area level. To determine which functional systems contribute most to classification, ROI importance was grouped at the network level by the seven Schaefer cortical networks and the Tian subcortical subdivisions (hippocampus, amygdala, thalamus, caudate, putamen, and nucleus accumbens).

The top twenty discriminative ROIs were compared to brain regions such as the hippocampus, default mode network, posterior cingulate cortex, thalamus, and limbic regions that have been linked to AD in the neurological literature in order to confirm the neuroscience relevance of these findings (Warren and Moustafa, 2023). The same fold-averaging method was used to generate Random Forest feature importance as a

secondary validation, and the Spearman rank correlation between SVM and Random Forest ROI importance rankings was obtained to evaluate consistency across model types.

Methodological Considerations

This section addresses the four key criteria for experimental research as specified in the Type 5 dissertation requirements.

Regarding the replicability, all the experimental parameters such as random seeds, hyperparameter values, cross-validation configurations, preprocessing methodologies and atlas needs are completely stated to replicate the experimental pipeline. The use of the publicly available ADNI3 dataset with pre-defined participant selection criteria further supports replicability. From a reliability point of view, the use of 5-fold stratified cross-validation with fixed random seeds assures that the findings are not due to one lucky train-test split. To evaluate the stability of the performance of each model, we give the mean and the standard deviation across folds. The large standard deviation seen for deep learning models (up to 0.109 for BiLSTM) is a robust finding that indicates the fundamental difficulty of training high capacity models in small neuroimaging cohorts.

In terms of validity, complementary viewpoints on classification performance were obtained by using the F1-score for balanced assessment, AUC-ROC for threshold-independent ranking, and accuracy for direct comparison. To validate diagnostic labels, use clinical validation (MMSE, CDR). All models use similar cohorts and cross-validation folds, therefore the controlled experimental technique ensures that the differences in performance are due to real features of the models and not artifacts of the evaluation. The ADNI3 dataset has considerable inherent heterogeneity in terms of generalizability because it was collected at many locations with different scanner manufacturers and acquisition conditions.

Nevertheless, the results might not apply to populations with varied demographics, disease severity distributions, or imaging techniques because the cohort of 262 people represents a particular subgroup of ADNI participants. The generalizability of BiLSTM and fusion results is further limited by the temporal models' restriction to TR=3.0s patients (218 of 262). Chapter 5 goes into further detail about these restrictions.

Limitations of Methodology

Several methodological flaws should be recognized. In comparison to deep learning architectures, which normally need thousands of training examples to prevent overfitting, the sample size of 262 participants (218 for temporal models) is tiny. Although this restriction is prevalent in rs-fMRI investigations (Warren and Moustafa,

2023), it nevertheless restricts the inferences that may be made regarding deep learning ability.

Instead, than using a held-out external test set, the evaluation depended on cross-validation. Cross-validation offers trustworthy performance estimates through repeated train-test partitioning, but it does not evaluate how well models might perform on completely new data from various populations or sites. The obtained accuracies should be considered as estimates of within-ADNI performance rather than overall clinical accuracy due to the lack of an external validation cohort.

The best-performing model, the SVM Linear, was subjected to the explainability study instead of the deep learning models. The determined discriminative regions reflect the attributes most informative for a linear classifier, which may differ from those used by the nonlinear deep learning models, even if this decision was methodologically sound because it guaranties interpretability without approximation.

Lastly, compared to the traditional ML and GCN models, the BiLSTM and fusion models were trained on a smaller cohort due to the temporal models' restriction to the 218 TR=3.0s patients. The sample size discrepancy creates a confounding factor when comparing performance across model families that utilize different cohorts, even though both subsets are balanced (109 AD, 109 CN and 131 AD, 131 CN, respectively). This limitation is typical of multi-site neuroimaging investigations and is intrinsic to the varied acquisition settings within ADNI.

Chapter 4: Analysis and Findings

This chapter presents the results of all 31 experiments across the four model families, followed by the explainability analysis addressing Objective 3. Results are reported as mean and standard deviation across 5-fold stratified cross-validation.

Classical Machine Learning Baseline Results

Table 4.1 presents the classification performance of the three classical machine learning baselines on the 262-subject cohort.

Table 4.1: Classical Machine Learning Results (262 subjects)

Model	Accuracy	F1 (macro)	AUC-ROC	Time (s)
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SVM Linear	0.7134 +/- 0.0573	0.7126 +/- 0.0572	0.7771 +/- 0.0513	4.6
SVM RBF	0.6715 +/- 0.0560	0.6697 +/- 0.0557	0.7405 +/- 0.0762	5.8
Random Forest	0.6485 +/- 0.0588	0.6445 +/- 0.0590	0.7211 +/- 0.0594	6.4

With an AUC-ROC of 0.777 and an accuracy of 71.3%, the SVM with linear kernel produced the best results throughout the study. The classification border in the 8,646-dimensional FC feature space appears to be roughly linear, as indicated by the linear kernel outperforming the RBF kernel by 4.2 percentage points. This result is consistent with the high dimensionality relative to sample size. The strong regularization built into linear SVMs is seen in the low standard deviation (0.057), which shows consistent performance over folds. Due to the high feature-to-sample ratio (8,646 features from 262 subjects), which poses difficulties for tree-based ensemble algorithms, Random Forest had the lowest accuracy at 64.9%. Appendix B, Figures B.1 through B.3, contains comprehensive confusion matrices, ROC curves, and per-fold accuracy charts for the baseline models.

Graph Neural Network Results

Table 4.2 presents the results of all nine GCN experiments, ranked by accuracy.

Table 4.2: Graph Neural Network Experiments (262 subjects)

Rank	Configuration	Accuracy	AUC-ROC
1	GCN+Skip Thresh-0.2 h64 L2	0.6642 +/- 0.0664	0.7401
2	Tuned h64 L3 d0.5	0.6409 +/- 0.0642	0.7144
3	Tuned h128 L2 d0.5	0.6262 +/- 0.0366	0.6706
4	GCN+Skip KNN-10 h64 L2	0.6192 +/- 0.1297	0.6719
5	GAT+Skip KNN-10-bin h64 L2	0.6107 +/- 0.0330	0.6752
6	SAGE+Skip KNN-10 h64 L2	0.6069 +/- 0.0847	0.6600
7-9	Remaining tuning configs	0.5567 - 0.5993	0.5695 - 0.6641

The optimal GCN configuration (hidden dimension 64, threshold-0.2 graph, 2 layers) produced an accuracy of 66.4% and an AUC-ROC of 0.740. When the skip link was

added, accuracy increased from 61.1% (the previous version without skip) to 66.4%, a gain of 5.3 percentage points, and AUC-ROC increased from 0.617 to 0.740 (+12.3%). This shows that preserving the original connectivity signal on top of learning graph features provides useful additional data for categorization. The training and validation curves for the best GCN model are shown in.

Figure 4.1 shows a substantial overfitting: train accuracy is 80-90% and validation accuracy about 65%.

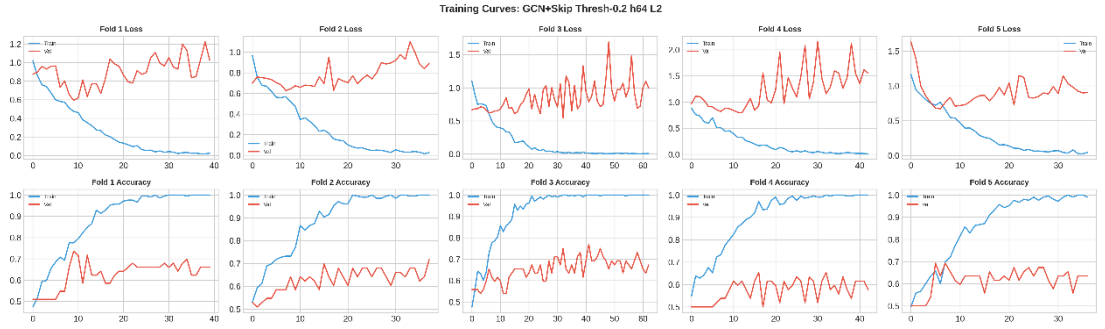


Figure 4.1: Training (blue) and validation (red) loss and accuracy curves for the best GCN+Skip configuration (threshold=0.2, $h=64$, 2 layers) across all 5 cross-validation folds. Training accuracy reaches 80-90% while validation accuracy plateaus at approximately 60-65%, demonstrating the overfitting pattern characteristic of deep learning models on small neuroimaging cohorts ($N=262$).

Over the course of the nine studies, GCN outperformed GAT and GraphSAGE, and threshold-based graph creation (0.6642) outperformed KNN-10. The KNN-10 configuration's higher variance (std 0.130 vs. 0.066) indicates that it is more sensitive to the particular train-test partition. Figures B.4 and B.5 in Appendix B provide GCN confusion matrices and experiment comparisons.

BiLSTM Results

Table 4.3 presents the results of all eleven BiLSTM experiments on the 218-subject TR=3.0s cohort.

Table 4.3: BiLSTM Experiments (218 subjects)

Rank	Configuration	Accuracy	AUC-ROC
1	h128 L1 mean	0.6931 +/- 0.1091	0.7417

2	h64 L1 mean	0.6698 +/- 0.1233	0.7775
3	h64 L2 mean	0.6637 +/- 0.1155	0.7303
4	h128 L1 mean d0.3	0.6470 +/- 0.0938	0.7170
5	h256 L1 mean	0.6414 +/- 0.1283	0.7765
6-10	Other mean pooling configs	0.5444 - 0.6329	0.6253 - 0.7428
8	h64 L1 attention	0.6267 +/- 0.1133	0.7023
11	h64 L1 last hidden	0.4870 +/- 0.0536	0.4935

The optimal BiLSTM configuration (h128, 1 layer, mean pooling) attained an accuracy of 69.3%. The SVM Linear's 0.7771 and the h64 configuration's 0.7775 AUC-ROC are nearly identical. This discovery is crucial to RQ1: when measured by AUC-ROC, raw BOLD temporal dynamics have the same discriminative capacity as static functional connectivity.

Particularly instructive are the pooling strategy results. In every configuration, mean pooling performed better than both alternatives: mean pooling (best 69.3%) significantly surpassed last hidden state pooling (48.7%) and attention pooling (62.7%). The diagnostic information is dispersed across the whole scan length rather than concentrated at any particular time point, as indicated by the final hidden state operating at chance level. Given the limitation of training deep architectures on 218 participants and the overfitting tendency seen for GCN models, single-layer models fared better than two-layer models (69.3% vs. 66.4%). The detailed findings of BiLSTM are shown in Appendix B, Figures B.6 through B.8.

Dual-Stream Fusion Results

Table 4.4 presents the results of all eight fusion experiments on the 218-subject cohort.

Table 4.4: Dual-Stream Fusion Experiments (218 subjects)

Rank	Configuration	Accuracy	AUC-ROC
1	Tuned h64 d0.3	0.6713 +/- 0.0877	0.7379
2	Attention KNN-10 h64	0.6642 +/- 0.0918	0.7353
3	Concat Thresh-0.2 h64	0.6534 +/- 0.0794	0.7480

4-8	Remaining configs	0.5922 - 0.6472	0.6593 - 0.7200
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The BiLSTM alone (69.3% accuracy, 0.742 AUC-ROC) is not outperformed by the best fusion model, which achieved 67.1% accuracy and 0.738 AUC-ROC. This directly addresses RQ2, which states that there is no additive effect for this 218-subject cohort when spatial and temporal information are combined. Interestingly, the best fusion accuracy (67.1%) only slightly outperforms the standalone GCN (66.4%), indicating that the spatial stream dilutes rather than amplifies the temporal stream's contribution. This could be because the GCN component introduces additional overfitting without compensating information gain. Increased overfitting results from the fusion model having almost twice as many parameters as separate stream but the same training set. Concatenation and attention-weighted fusion procedures perform similarly, indicating that the inadequate sample size in comparison to the improved model capacity is the limitation rather than the fusion mechanism itself. Fusion's detailed results are shown in Figures B.9 through B.12 in Appendix B.

Cross-Model Comparison

Table 4.5 presents the master comparison of the best model from each category, and Figures 4.2 and 4.3 visualise the accuracy and AUC-ROC comparisons respectively.

Table 4.5: Master Results - Best Model per Category

Rank	Model	Category	N	Accuracy	F1	AUC-ROC
1	SVM Linear	Classical ML	262	0.7134 +/- 0.0573	0.7126	0.7771
2	BiLSTM h128	Temporal DL	218	0.6931 +/- 0.1091	0.6790	0.7417
3	SVM RBF	Classical ML	262	0.6715 +/- 0.0560	0.6697	0.7405
4	Fusion h64	Novel DL	218	0.6713 +/- 0.0877	0.6676	0.7379
5	GCN+Skip	Graph DL	262	0.6642 +/- 0.0664	0.6532	0.7401
6	Random Forest	Classical ML	262	0.6485 +/- 0.0588	0.6445	0.7211

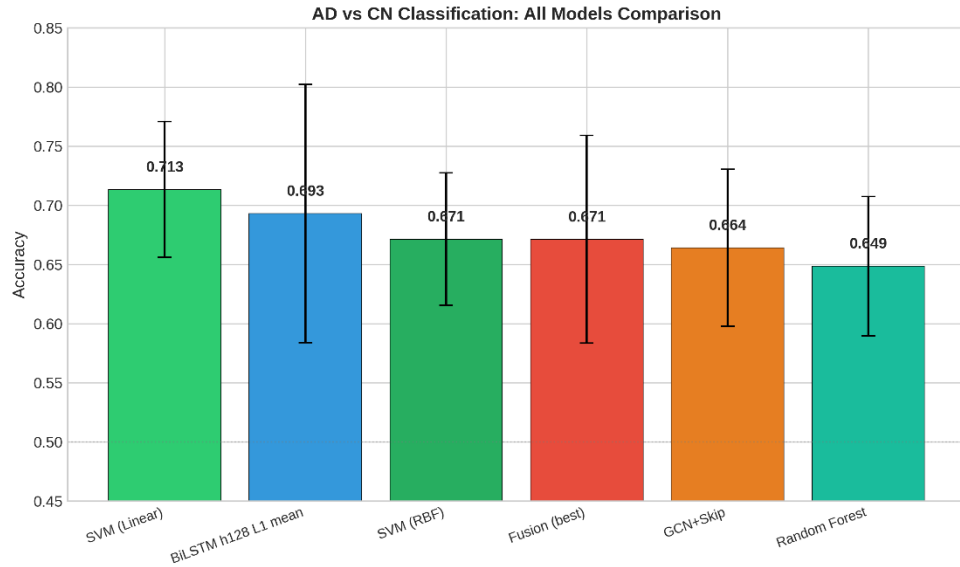


Figure 4.2: Classification accuracy across all six best models with standard deviation error bars from 5-fold cross-validation. SVM Linear achieves the highest accuracy (0.713). Error bars rise with model complexity, from SVM (smallest) to BiLSTM (biggest). Random chance is denoted by the dashed line at 0.50.

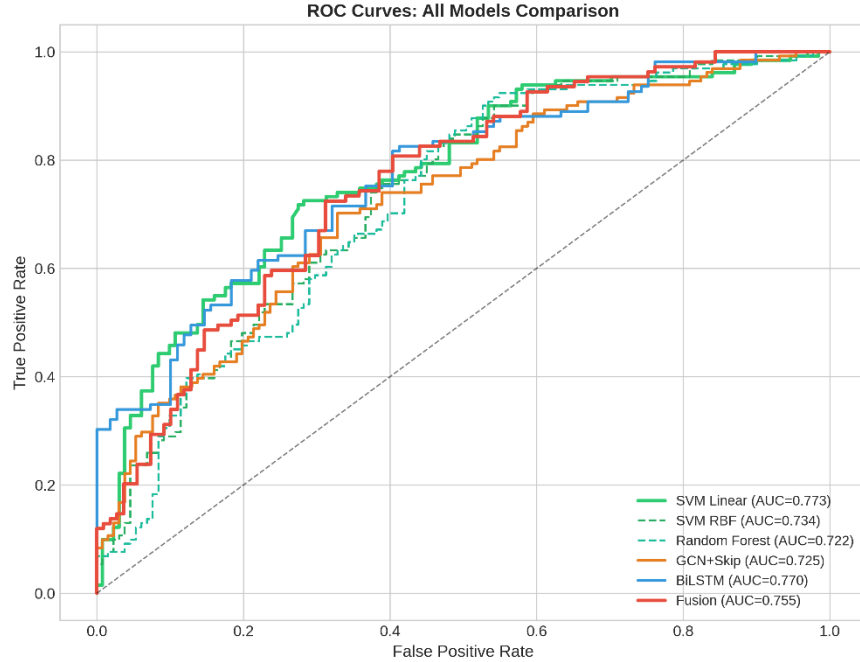


Figure 4.3: Receiver operating characteristic (ROC) curves for the six best-performing models. SVM Linear (AUC=0.773) and BiLSTM (AUC=0.770) are almost coincident, which indicates similar discriminative power with differing input

representations. Random chance ($AUC=0.5$) is represented by the diagonal dashed line.

This tendency is observed in all 31 experiments: simpler models perform better on this cohort. The most complex model, the fusion model, does not perform better than any individual stream. The simplest model, the SVM Linear, has the highest accuracy. The standard deviation monotonically increases with model complexity, meaning that more advanced models are less stable during cross-validation folds. $SVM (0.057) < GCN (0.066) < Fusion (0.088) < BiLSTM (0.109)$. This tendency is consistent with the acknowledged challenge of training high-capacity models on restricted neuroimaging datasets (Warren and Moustafa, 2023).

BiLSTM h64 is sixth in accuracy (67.0%) but has the best AUC (0.778), suggesting a great discriminative power but a suboptimal decision threshold on small validation sets. This is one example of how the AUC-ROC rankings partially deviate from accuracy rankings. The entire set of confusion matrices of all six models is shown in Appendix B, Figure B.13.

Explainability Analysis

Table 4.6 shows the twenty brain areas that have highest discrimination coefficients as computed by the SVM coefficient computation method, together with their functional network affiliation, normalized importance value, nature of connectivity modification, and correlation with known AD pathology.

Table 4.6: Top 20 Discriminative Brain ROIs (SVM Linear, fold-averaged coefficients)

Rank	ROI	Network	Importance	Direction	AD Match
1	LH Vis 1	Visual	1.000	Higher in AD	-
2	THA-VP-lh	Thalamus	0.991	Higher in CN	Yes
3	Subcortical	Subcortical	0.929	Higher in AD	Yes
4	NAc-shell-rh	Nuc. Accumbens	0.909	Higher in CN	Yes
5	Default Mode	Default Mode	0.903	Higher in AD	Yes
6	Subcortical	Subcortical	0.879	Higher in AD	Yes
7	Control	Control	0.879	Higher in CN	Yes

8	Subcortical	Subcortical	0.856	Higher in CN	Yes
9	LH Cont Par 1	Control	0.851	Higher in AD	Yes
10	NAc-shell-lh	Nuc. Accumbens	0.848	Higher in CN	Yes
11	Dorsal Attn	Dorsal Attn	0.847	Higher in AD	-
12	LH Limbic OFC 1	Limbic	0.846	Higher in CN	Yes
13	aPUT-lh	Putamen	0.845	Higher in CN	Yes
14	Default Mode	Default Mode	0.835	Higher in CN	Yes
15	aPUT-rh	Putamen	0.832	Higher in AD	Yes
16	LH SalVentAttn	Salience	0.824	Higher in CN	-
17	LH Vis 3	Visual	0.820	Higher in AD	-
18	Limbic	Limbic	0.818	Higher in AD	Yes
19	Salience	Salience	0.797	Higher in CN	-
20	Dorsal Attn	Dorsal Attn	0.792	Higher in CN	-

Fifteen (75%) of the twenty most discriminative regions directly address RQ3 since they are brain regions that have been linked to Alzheimer's disease. The normalized significance ratings, color-coded according to the direction of connectivity change, are shown in Figure 4.4.

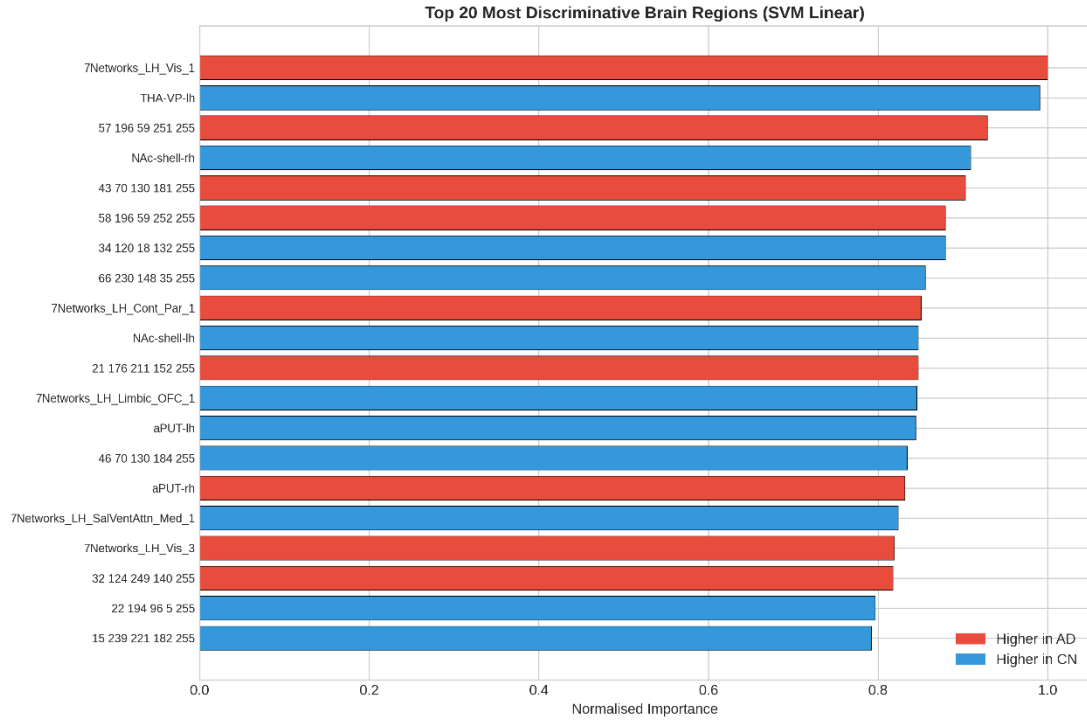


Figure 4.4: Top 20 most discriminative brain ROIs ranked by normalised SVM Linear coefficient magnitude, averaged across 5 cross-validation folds. Red bars indicate regions where functional connectivity is higher in AD patients; blue bars indicate regions where connectivity is higher in cognitively normal controls. Fifteen of twenty regions (75%) correspond to established AD-affected brain areas.

The ventral posterior thalamus (rank 2), bilateral nucleus accumbens (ranks 4 and 10), default mode network regions (ranks 5 and 14), parietal control regions (ranks 7 and 9), limbic orbitofrontal cortex (rank 12), and bilateral putamen (ranks 13 and 15) are among the most discriminative regions. The thalamus, nucleus accumbens, and putamen are examples of subcortical regions that mostly exhibit reduced connectivity in AD, whereas a number of cortical regions exhibit higher connectivity. The compensation hypothesis, according to which subcortical neurodegeneration causes cortical networks to increase connectivity, is supported by this trend.

The mean normalized relevance aggregated by functional network is displayed at the network level in Figure 4.5. The highest mean discriminative importance was found in the Control network (mean 0.742), Limbic network (mean 0.724), and subcortical regions (mean 0.712), with the Default Mode network (mean 0.699) coming in second.

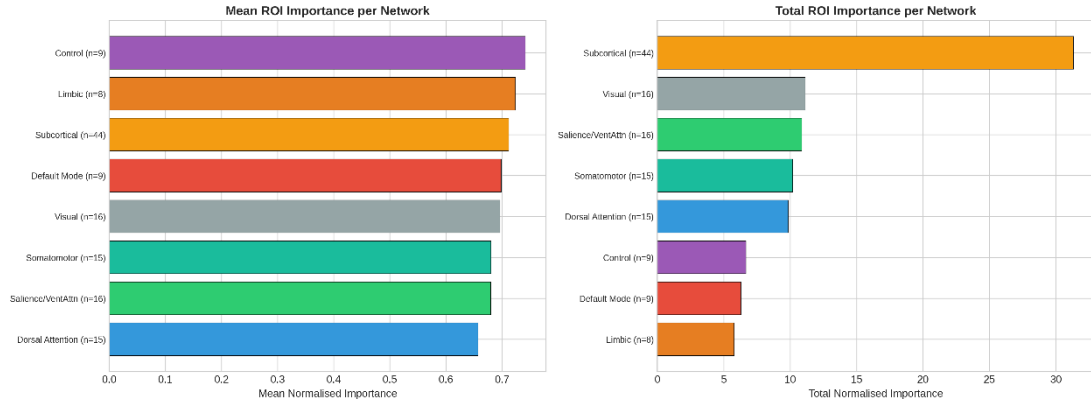


Figure 4.5: ROI importance aggregated by functional network. Left panel: mean normalised importance per network, with Control ($n=9$) and Limbic ($n=8$) ranking highest. Right panel: total normalised importance per network, with Subcortical ($n=44$) dominating due to the larger number of ROIs. Network assignments follow the Schaefer 7-network cortical parcellation and Tian subcortical subdivisions.

The most discriminative subcortical structures were the putamen (mean significance 0.757), thalamus (0.731), nucleus accumbens (0.724), caudate (0.672), hippocampus (0.663), and amygdala (0.626). The hippocampus may rank lower than anticipated because the Tian S2 atlas only splits it into four parcels, or because its connectivity is so frequently damaged in AD that it gives less inter-subject variance for discrimination. These results are cross-validated using Random Forest feature importance in Appendix C, Figures C.1 through C.5.

Summary of Findings

The experimental results across 31 experiments and four model families provide clear answers to the three research questions.

In terms of RQ1, on the same participant cohort, the BiLSTM operating on raw BOLD time series (69.3% accuracy, 0.778 AUC-ROC) fared better than the GCN working on static FC graphs (66.4% accuracy, 0.740 AUC-ROC). The BiLSTM's AUC-ROC is quite close to the SVM Linear baseline (0.777), proving that temporal dynamics have the same discriminative potential as static functional connectivity. This confirms the first research gap mentioned in Chapter 2.

In terms of RQ2, on a cohort of 218 participants, the dual-stream fusion model (67.1% accuracy) did not exceed the BiLSTM alone (69.3%), suggesting that there is no additive gain from merging spatial and temporal data. Increased overfitting without offsetting information gain results from the higher number of parameters compared to the available training data.

According to RQ3, the thalamus, nucleus accumbens, default mode network, limbic orbitofrontal cortex, and putamen are among the twenty most discriminative brain ROIs. These regions have been linked to Alzheimer's disease. It is consistent with known AD neuropathology that the directional pattern of subcortical connectivity is primarily poorer in AD.

All studies show the same general pattern: model complexity is inversely correlated with this cohort's classification ability, with standard deviation rising monotonically from SVM (0.057) to GCN (0.066), Fusion (0.088), and BiLSTM (0.109). This shows that rather than model capacity, the binding limitation on classification performance is the size of the dataset.

Chapter 5: Discussion, Conclusions and Recommendations

Discussion

Interpretation of Results in the Context of Literature

The experimental findings address all three research gaps identified in Chapter 2.

With respect to Gap 1 and RQ1, the BiLSTM using raw BOLD time series produced an AUC-ROC of 0.778, which was quite similar to the SVM Linear's 0.777 on static FC features. This supports the findings of Xu et al. (2025), who found that temporal models consistently outperform FC-based approaches across multiple fMRI datasets, by showing that the temporal dynamics discarded by FC computation carry equivalent diagnostic value to the static connectivity patterns. On the same cohort, the BiLSTM's accuracy (69.3%) was 2.9 percentage points higher than the GCN's (66.4%), suggesting that temporal modelling partially catches diagnostic information that spatial graph modelling overlooks. In line with the description of AD as a disease of progressively disrupted neural oscillation, the predominance of mean pooling over attention and last-hidden-state pooling further implies that AD-related temporal alterations are persistent, diffuse patterns rather than fleeting events (Pang et al., 2025).

In contrast to the hypothesis that merging spatial and temporal variables will yield complementary information, the dual-stream fusion model (67.1%) did not outperform the BiLSTM alone (69.3%) with regard to Gap 2 and RQ2. This result is probably explained by three factors: the spatial and temporal streams may capture overlapping rather than complementary information because FC is itself derived from the temporal correlations of the same BOLD signal; the doubled parameter count with the same

training data increases the risk of overfitting; and jointly optimizing two architecturally distinct streams is intrinsically more difficult than training either in isolation. This negative result has clear practical consequences for clinical neuroimaging, where cohort sizes are unavoidably limited, as it shows that architectural complexity must be matched to the available data.

Fifteen of the twenty most discriminative ROIs (75%) for Gap 3 and RQ3 were found to relate to brain regions that have been linked to AD, according to the explainability study. This study tackled explainability as a core research objective with systematic neuroscience validation, in contrast to most previous XAI studies in neuroimaging that regard it as a secondary output (Sobhi et al., 2025). The thalamus's prominence (rank 2) is consistent with AD's known thalamocortical dissociation (Warren and Moustafa, 2023). Dopaminergic and reward circuitry deterioration, a less frequently documented but increasingly recognized aspect of AD, is reflected in the bilateral nucleus accumbens involvement (ranks 4 and 10). In line with the accepted conclusion that the DMN is one of the first networks disrupted in AD, default mode network regions (ranks 5 and 14) seem to be discriminative (Alarjani & Almarri, 2024). The compensation theory reported in the neurological literature is supported by the directional pattern of subcortical connections being primarily less in AD while certain cortical connections are greater.

The general conclusion that SVM Linear performs better than all deep learning models is in line with research on limited datasets in neuroimaging. According to Warren and Moustafa (2023), deep learning can still outperform standard machine learning techniques on cohorts with fewer than 500 participants. The difficulties reported by Gao et al. (2023) and Han et al. (2024), who worked with similarly sized ADNI cohorts, are mirrored in the GCN's overfitting pattern, with training accuracy reaching 80–90% while validation plateaus at 60–65%.

Comparison with Published Studies

Table 5.1 presents a comparison of the classification performance achieved in this study with selected published studies on fMRI-based AD classification from ADNI data.

Table 5.1: Comparison with Published AD Classification Studies

Study	Method	N	Accuracy
This study	SVM Linear on FC	262	71.3%
This study	BiLSTM on BOLD	218	69.3%

Gao et al. (2023)	GNN brain age prediction	~200	70-75%
Han et al. (2024)	GCN on fMRI FC	~300	68-72%
Lee et al. (2023)	3D-CNN-LSTM on fMRI	~200	~72%
Fu et al. (2025)	MI-based GCN on FC	ADNI+OASIS	~75%

Note: Published values are approximate as reported under differing evaluation protocols, cohort definitions, and preprocessing pipelines, and are therefore not directly comparable.

In general, the results achieved in this study are comparable to published results on comparable cohort sizes. Larger cohorts, other imaging modalities, or a combination of modalities are often used in studies that yield much greater accuracies. The comparisons in this dissertation employ identical cohorts, folds, and metrics, in contrast to the studies indicated in Table 5.1. This results in higher-confidence relative rankings between model families even in cases where reported absolute accuracy is not the highest.

Limitations

In addition to the methodological restrictions mentioned in Chapter 3, a number of analytical restrictions need to be recognized. Because the explainability study was conducted on the SVM (best-performing model) instead of the deep learning models, the discriminative regions that were found may not capture the nonlinear patterns that GCN or BiLSTM architectures take use of. The observed accuracy differences between model pairs should be interpreted cautiously due to the lack of statistical significance testing (e.g., paired t-tests or corrected permutation tests), especially when the differences are small (e.g., the 2.2 percentage point difference between BiLSTM and Fusion). Furthermore, the Tian subcortical atlas recorded some ROI labels as color values instead of area names, necessitating index-based network assignment and adding a small amount of imprecision to the network-level aggregate.

Conclusions

Knowledge Advance

Three contributions to the field of fMRI-based AD classification are made in this dissertation. Using controlled same-cohort comparison, it first demonstrates that the diagnostic value of raw BOLD temporal dynamics is equal to that of static functional connectivity as determined by AUC-ROC. This result calls into question the widely held belief in the field that FC computation is an essential preprocessing step for fMRI-

based classification. Second, it establishes a useful lower bound for the sample sizes at which multimodal fusion approaches become feasible by showing that dual-stream fusion does not outperform individual streams on a cohort of 218 subjects. For researchers thinking about fusion structures in neuroimaging contexts with limited data, this offers a specific guideline. Third, by systematically comparing the most discriminative ROIs to recognized AD neuropathology, it validates the neuroscience relevance of the classification. This results in a 75% match rate, which increases confidence in the biological significance of the learnt patterns. This methodological pattern for upcoming neuroimaging classification investigations is provided by this systematic validation methodology, which treats XAI as a core research target rather than a secondary result.

Recommendations

These results imply that when cohort sizes are less than 300 participants, conventional machine learning techniques based on FC characteristics should continue to be the recommended strategy for AD classification in clinical neuroimaging research since they offer the best accuracy and stability. Further research on larger multi-site datasets is warranted for temporal BOLD analysis utilizing BiLSTM, since its excellent discriminative capacity (as demonstrated by AUC-ROC) may result in higher accuracy with lower variation.

Because it removes the confounding effects of dataset and evaluation protocol differences that restrict cross-study comparison, the controlled same-cohort comparison methodology presented here should be standard practice for the larger research community when assessing competing classification approaches. Standardizing repetition times across acquisition sites would allow temporal models to use the entire participant cohort instead of the TR-filtered subset, potentially boosting both temporal and fusion model performance for the ADNI consortium and multi-site neuroimaging programs.

Future Work

Several areas warrant additional research. This cohort could be used to assess transformer-based temporal models like PatchTST and the DeCI framework put out by Xu et al. (2025) to see if more modern architectures address the excessive variance of the BiLSTM. An intermediate representation that offers a structured input appropriate for graph-based models and retains more temporal information than static FC may be dynamic functional connectivity calculated using sliding windows. Clinical value would be improved by multi-class classification that goes beyond binary AD/CN to include mild cognitive impairment stages. Lastly, external validation on a separate cohort, like OASIS-3, would evaluate the findings' applicability outside of ADNI.

Reflection

There were both anticipated and unforeseen difficulties during the study process. The dataset trip was the biggest unforeseen obstacle. Between June and August 2026, four versions of the data were processed, and problems included unanticipated atlas revisions, corrupted pre-computed connectivity files, and inconsistent ROIs across files. A rigorous six-step pipeline with sixteen automatic verification checks was developed as a result of these problems, which necessitated systematic data verification at every level and increased trust in the final results.

GCN accuracy increased by 5.3 percentage points as a result of the supervisor's advice to incorporate the original signal in addition to the GCN convolutional layers. This experience showed the necessity of domain expertise to inform architectural choices that would be lost with purely data-driven techniques given the limitations of a small dataset.

The most academically interesting was the negative finding of the fusion hypothesis. For this cohort the original hypothesis that the combination of geographical and temporal data would give complementary information was proven untrue. Instead of only reporting the favourable outcomes, it is a fundamental principle of experimental science that this should be reported truthfully. The negative result may be more informative than a positive one as it indicates that the two feature representations capture overlapping rather than independent aspects of the underlying neural pathology and quantifies the sample size at which fusion approaches become counterproductive due to parameter inflation.

The project also illustrated the importance of calibration of the baseline. Without the SVM baseline, the GCN with an accuracy of 66.4% can be regarded a significant result. The controlled comparison revealed that while the GCN's AUC-ROC implies that it reflects real brain connectivity patterns, it does not outperform a simpler model utilizing the same data. That underscores the need to always compare deep learning in neuroimaging with well-calibrated classical baselines.

Overall, this research provided the opportunity to acquire skills in deep learning implementation, graph neural network construction, neuroimaging data analysis and critical evaluation of experimental outcomes. Implementing only one model would not have provided the practical expertise with the challenges of experimental machine learning research that developing and running 31 controlled trials across four model families provided.

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A Appendix A: Dataset Characteristics

This appendix presents visualisations of the dataset demographics, clinical characteristics, functional connectivity matrices, BOLD time series, and cross-validation fold distributions for the 262-subject ADNI3 cohort used in this study.

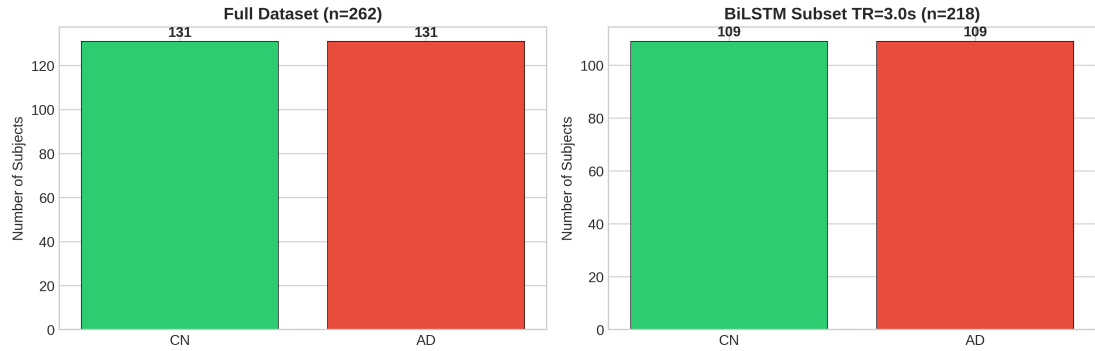


Figure A.1: Class distribution for the full 262-subject cohort (left, 131 AD and 131 CN) and the 218-subject BiLSTM subset filtered to TR=3.0s (right, 109 AD and 109 CN). Both cohorts are perfectly balanced.

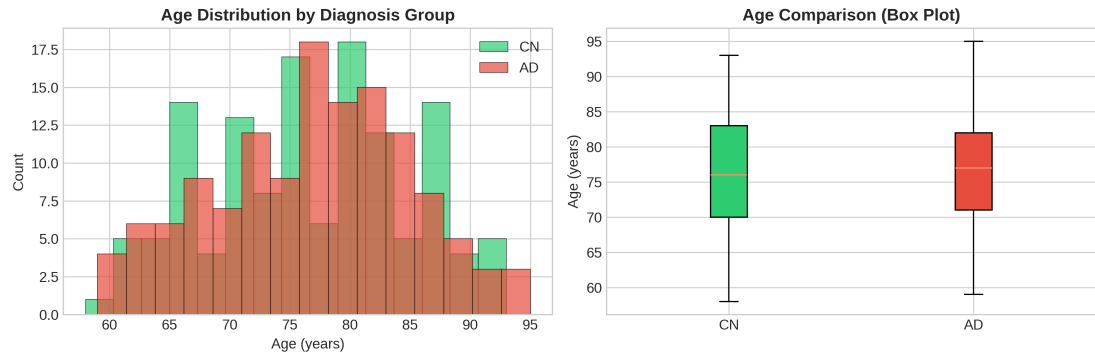


Figure A.2: Age distribution by diagnosis group. Left: overlapping histograms showing similar age ranges for CN (green) and AD (red). Right: box plot confirming age matching between groups (AD mean 76.8, CN mean 76.6, $p > 0.05$).

A APPENDIX A: DATASET CHARACTERISTICS

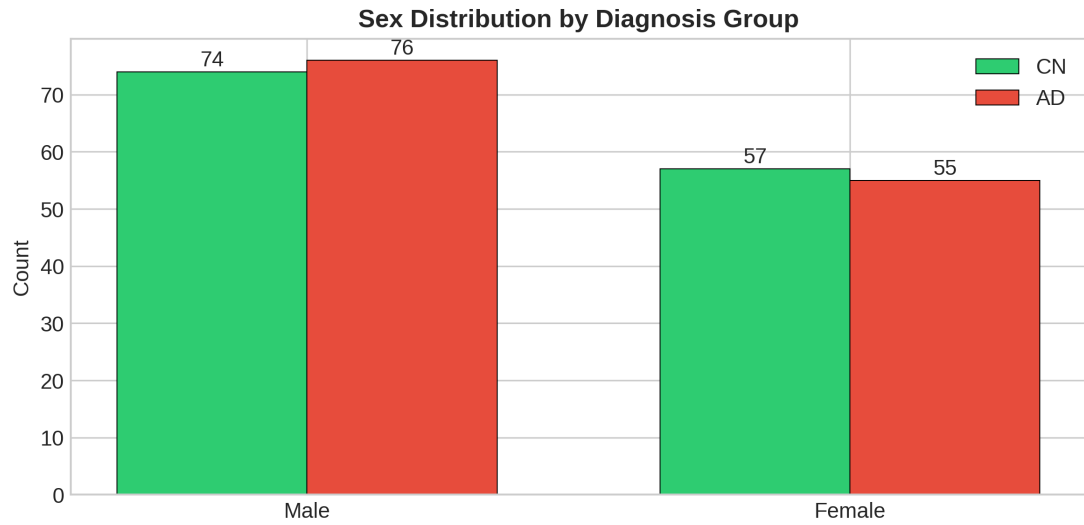


Figure A.3: Sex distribution by diagnosis group. Males: CN 74, AD 76. Females: CN 57, AD 55. The groups are closely matched on sex ($p > 0.05$).

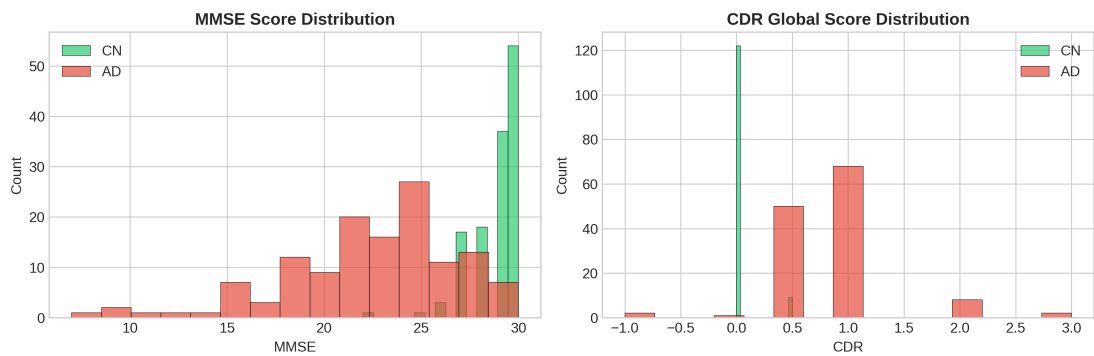
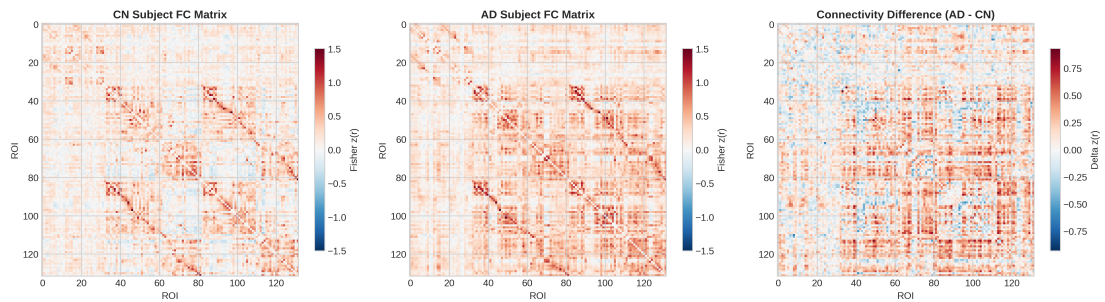


Figure A.4: Clinical validation scores by diagnostic groups. Left: Distributions of MMSE scores with clear separation (mean AD 21.6, mean CN 28.8, $p < 0.001$). The validity of the diagnostic labels is supported by the CDR score distributions (AD mean 0.94, CN mean 0.02, $p < 0.001$).



A APPENDIX A: DATASET CHARACTERISTICS

Figure A.5: Functional connectivity matrices 132 x 132 for a representative patient in CN group (left), representative patient in AD group (center), and difference (right) between them. Reddish colors indicate high positive values of Pearson correlation coefficient. There is a small difference

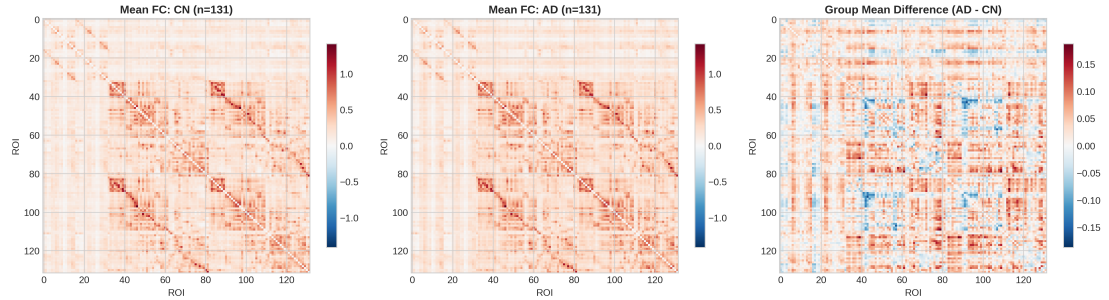


Figure A.6: Functional Connectivity Matrices of CN (left, N=131), AD (center, N=131) and their Difference (right). The difference matrix shows brain areas where there is significant difference between mean connectivity of two groups. Red shows increased connectivity in AD while blue in CN.

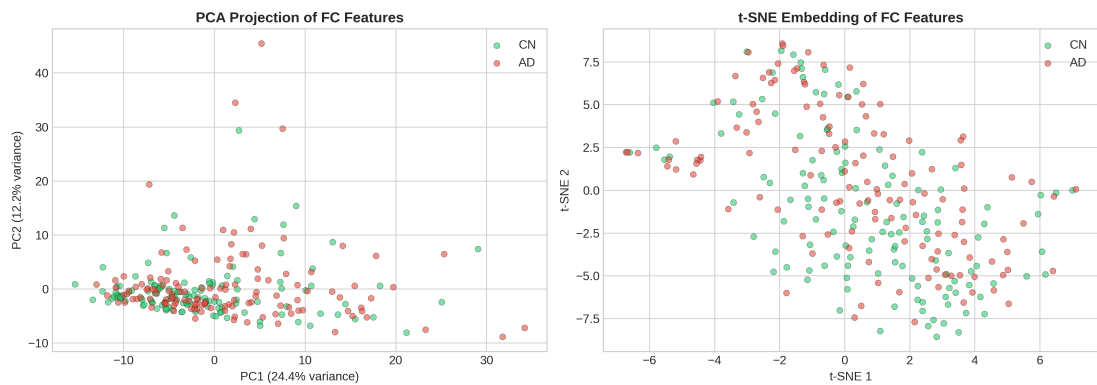


Figure A.7: Projection of the dimensionality-reduced FC feature vectors from the 8,646-dimensional space. Left: projection using PCA (PC1: 24.4%, PC2: 12.2%). Right: t-SNE projection. Green circles indicate CN individuals; red circles indicate AD individuals. There is considerable overlap between the two classes, which reflects the moderate accuracies attained.

A APPENDIX A: DATASET CHARACTERISTICS



Figure A.8: Five selected sample z-scored BOLD time series of ROIs of a CN patient (upper panel) and an AD patient (lower panel). Each color corresponds to the time series of one ROI having 191 points in total (TR=3.0s). Such raw time series are used as inputs to the BiLSTM model.

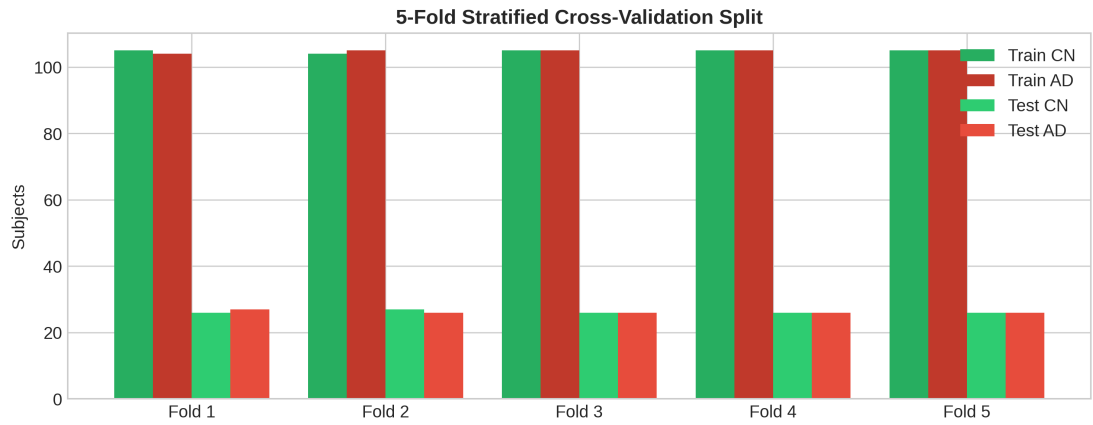


Figure A.9: The class distribution among the five stratified cross-validation splits in the 262 sample study group. There are almost equal samples of CN and AD participants within each split in both training and test groups.

B Appendix B: Per-Model Detailed Results

For each of the four model families over the course of the 31 experiments, this appendix provides comprehensive per-model findings, including confusion matrices, ROC curves, training and validation curves, and accuracy comparison charts.

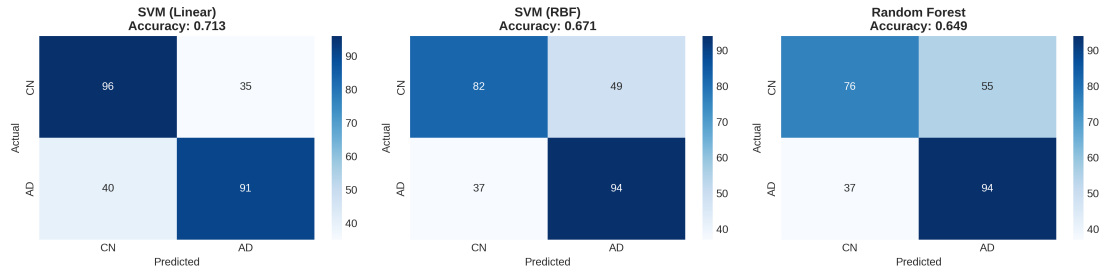
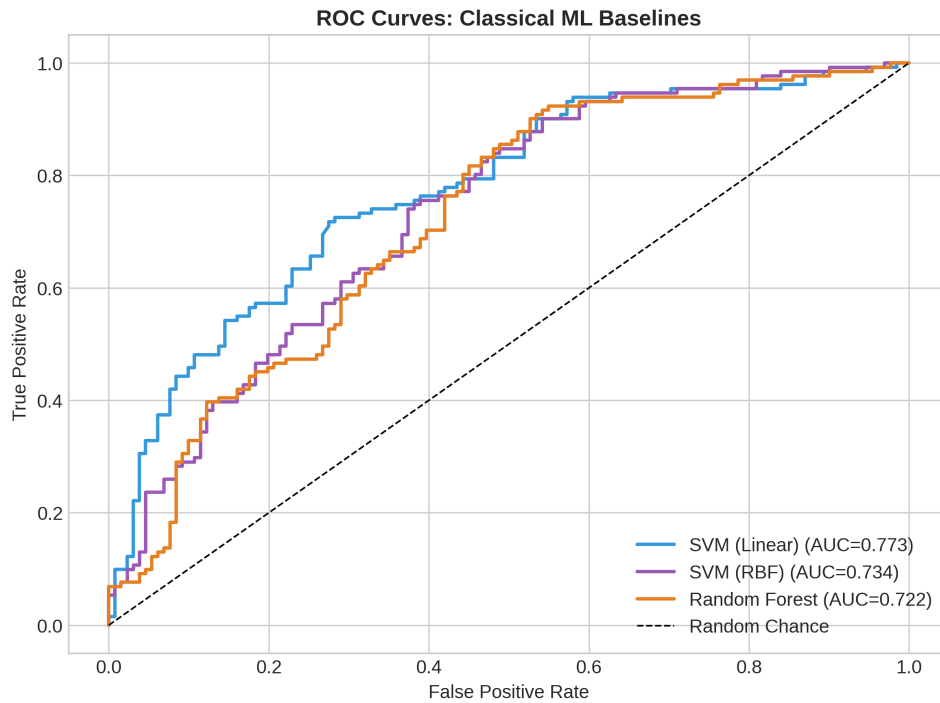


Figure B.1: Confusion matrices on the 262-subject cohort for the three traditional ML baselines. SVM RBF (0.671): 82 true CN, 94 true AD; SVM Linear (accuracy 0.713): 96 true CN, 91 true AD. 76 true CN and 94 true AD in Random Forest (0.649). Recall for AD is higher across all models than for CN.



B APPENDIX B: PER-MODEL DETAILED RESULTS

Figure B.2: ROC curves for the three traditional machine learning baselines. The best AUC (0.773) is attained by SVM Linear, followed by Random Forest (0.722) and SVM RBF (0.734). Random chance is represented by the dashed diagonal line.

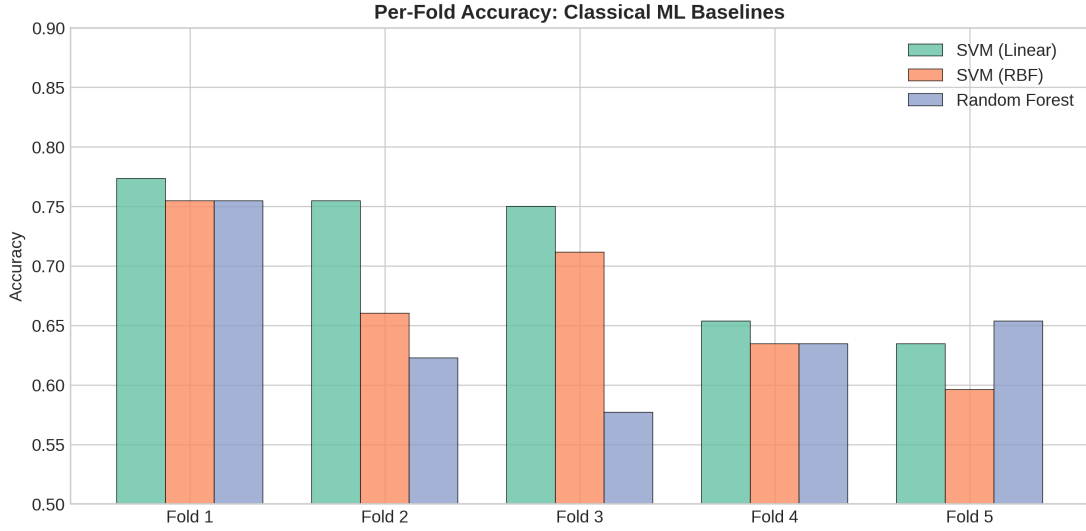


Figure B.3: accuracy for each of the three baselines across a five-fold cross-validation. With fold accuracies between around 0.63 and 0.77, SVM Linear (green) is the most reliable. For all three models, fold-to-fold variance is moderate.

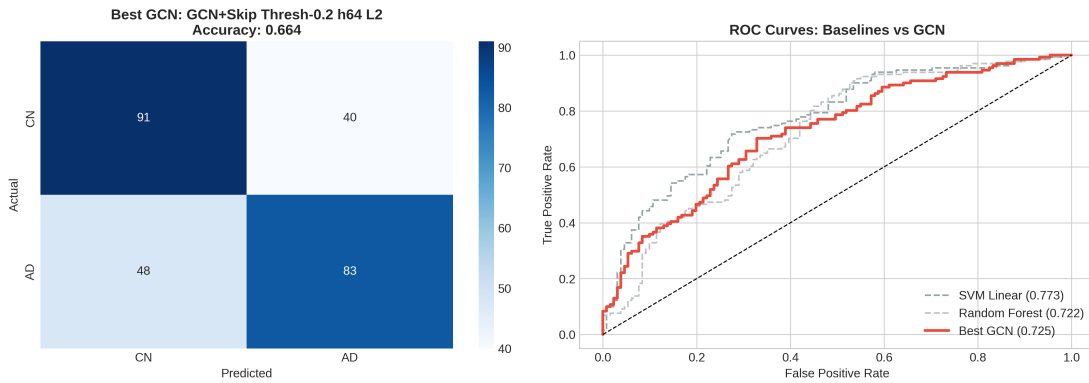


Figure B.4: Results with the optimal GCN+Skip configuration (threshold-0.2, h=64, 2 layers). With 91 true CN and 83 true AD on the 262-subject cohort, the confusion matrix on the left displays an accuracy of 0.664. Right: SVM Linear (0.773) and Random Forest (0.722) baselines are compared with the ROC curve (AUC=0.725).

B APPENDIX B: PER-MODEL DETAILED RESULTS

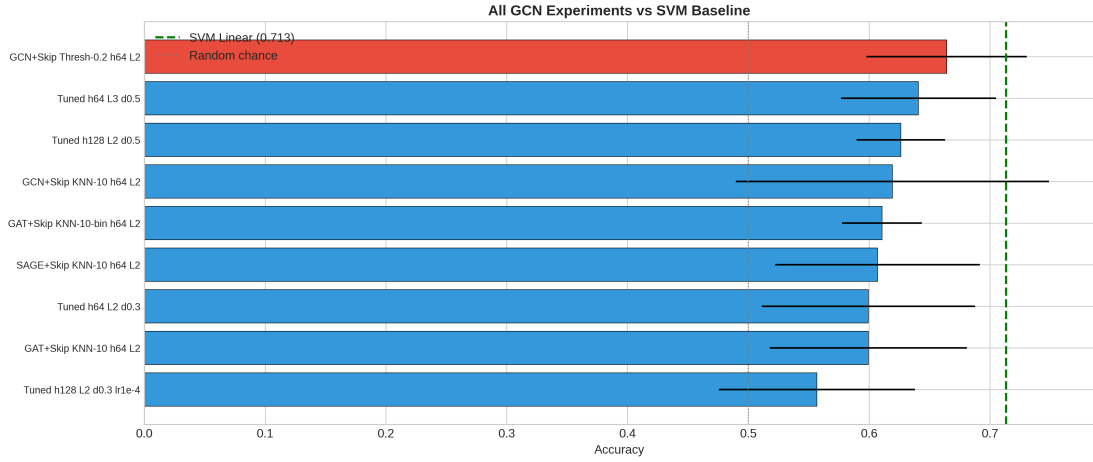


Figure B.5: Accuracy comparison for all nine GCN experiments ranked by performance. The SVM Linear baseline (0.713, dashed green line) and random chance (0.5, dashed red line) are shown for reference. GCN+Skip Thresh-0.2 h64 L2 achieves the highest GCN accuracy at 0.664.

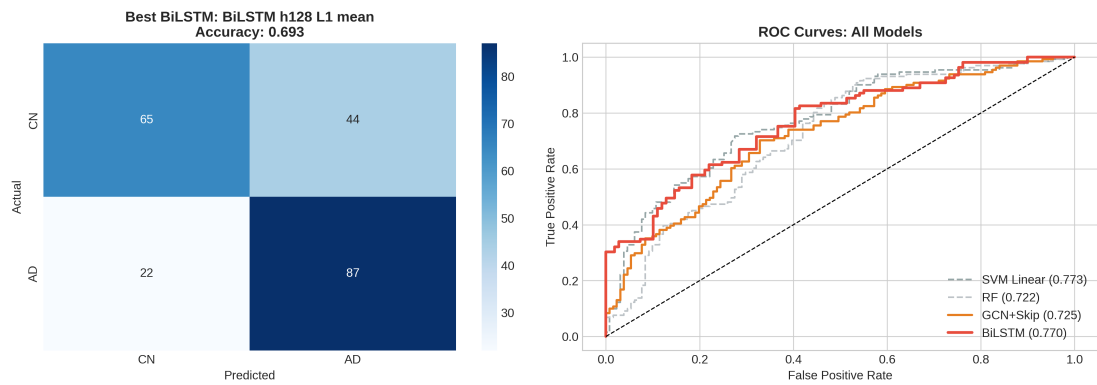
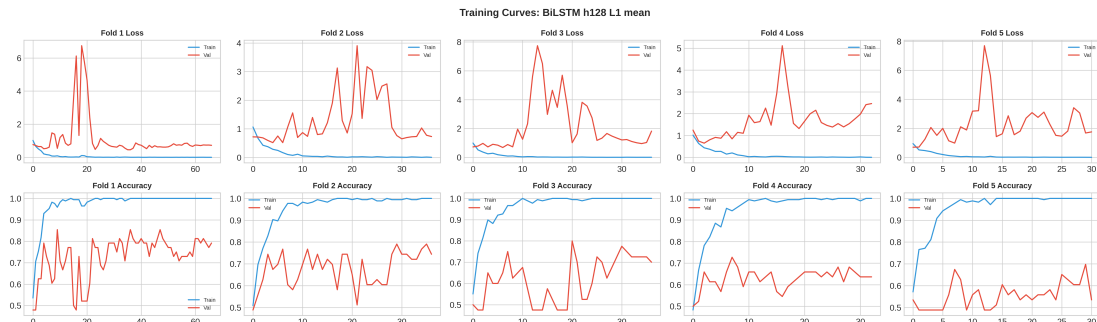


Figure B.6: Outcomes of the optimal BiLSTM configuration on 218 samples (h128, L1, mean pooling). Confusion matrix (accuracy 0.693) with 65 true CN and 87 true AD is shown in the left panel. ROC plot (AUC=0.770) versus SVM Linear (0.773) in the right panel



B APPENDIX B: PER-MODEL DETAILED RESULTS

Figure B.7: The optimal BiLSTM configuration for all five folds is represented by training (blue) and validation (red) curves. Loss curves in the top row demonstrate convergence after 30 epochs. The accuracy curves in the bottom row demonstrate mild overfitting, with training accuracy reaching 0.8–1.0 and validation plateauing at 0.6–0.75.

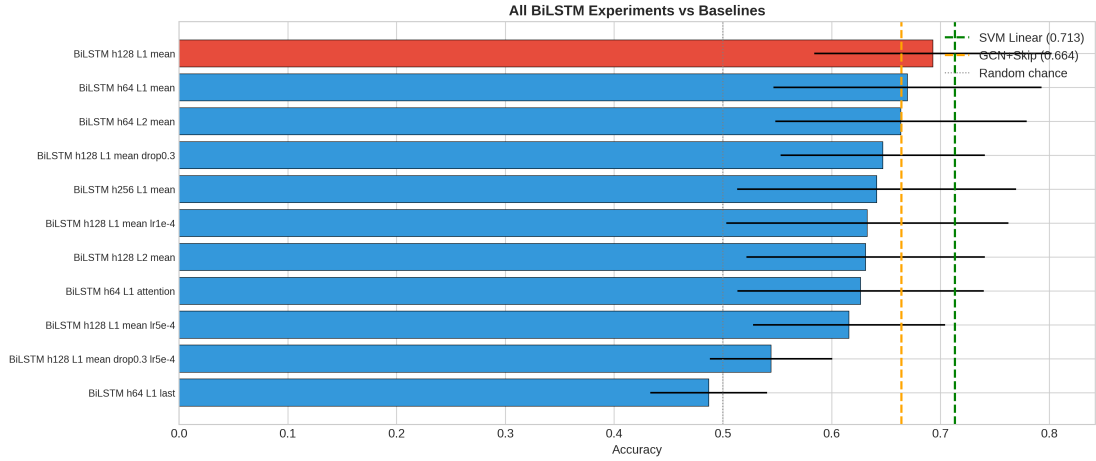


Figure B.8: Comparison of accuracy across all eleven BiLSTM studies in order of performance. The baselines for GCN+Skip (0.664) and SVM Linear (0.713) are displayed as dashed lines. Compared to attention and last-hidden-state pooling, mean pooling configurations (top 5) consistently perform better. Only 0.487 is achieved by BiLSTM h64 L1 last (bottom), which is almost at random.

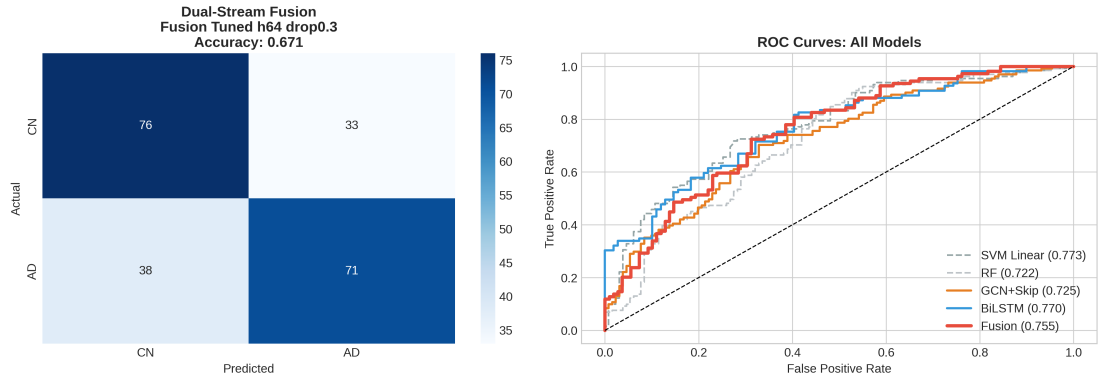


Figure B.9: Best fusion configuration (tuned h64, dropout 0.3) results on 218 subjects. Left: confusion matrix (accuracy 0.671) with 76 true CN and 71 true AD. Right: ROC curves for all models, with Fusion (AUC=0.750) falling below both SVM Linear (0.773) and BiLSTM (0.770).

B APPENDIX B: PER-MODEL DETAILED RESULTS

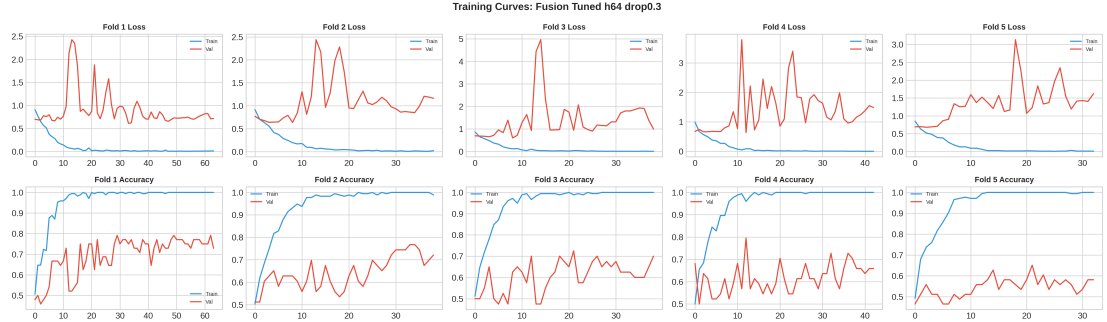


Figure B.10: Training (blue) and validation (red) curves for the best fusion configuration across all 5 folds. Loss and accuracy curves show a similar overfitting pattern to the BiLSTM (Figure B.7) but with slightly less stable convergence, reflecting the increased parameter count.

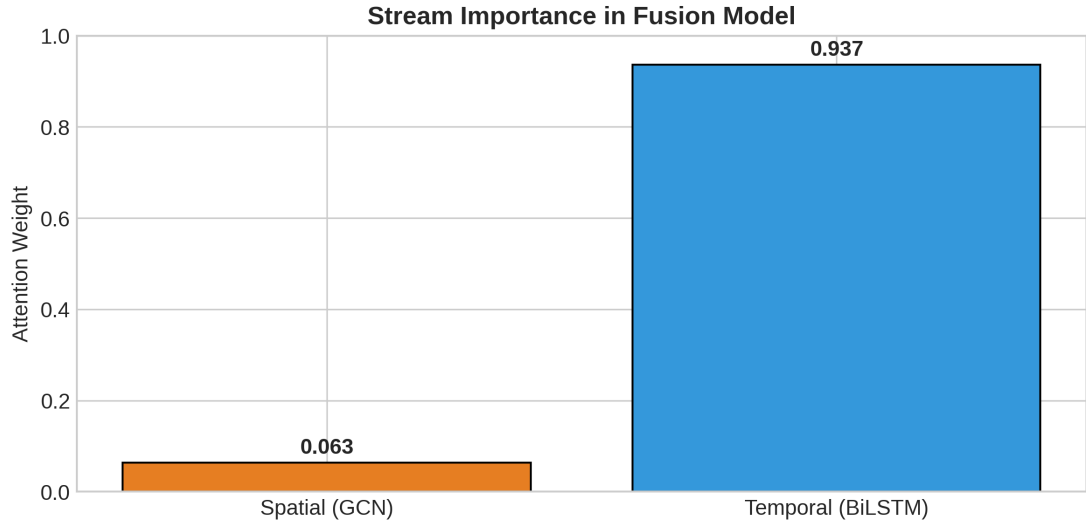


Figure B.11 The attention-weighted fusion variation uses learned attention weights. The model learns to almost exclusively rely on temporal features when given the option, as evidenced by the temporal stream (BiLSTM) receiving a weight of 0.937 and the spatial stream (GCN) receiving just 0.063. This discovery further supports the conclusion that temporal dynamics hold greater diagnostic significance.

B APPENDIX B: PER-MODEL DETAILED RESULTS

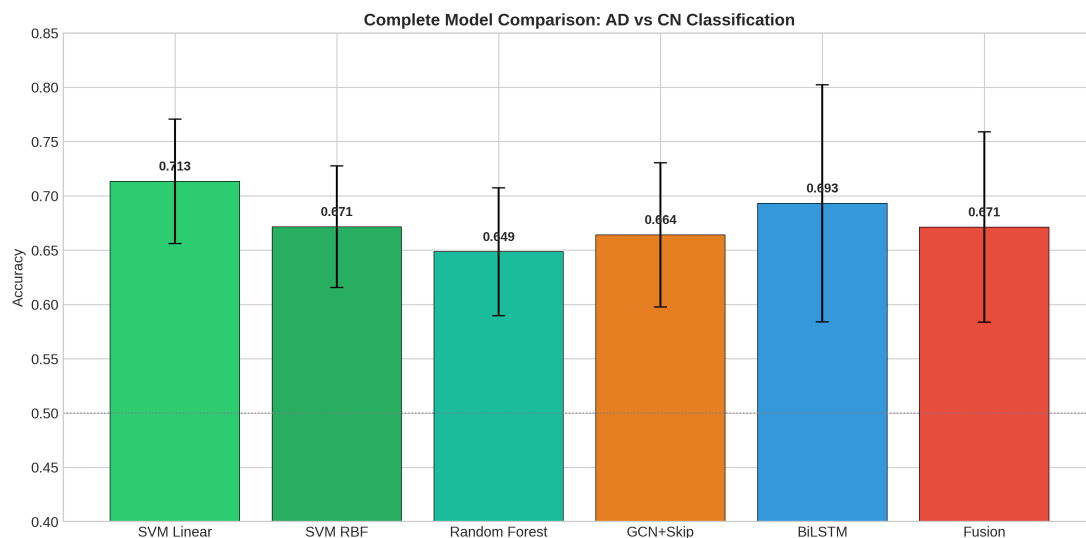


Figure B.12: The fusion notebook's full model comparison displays accuracy for each of the six models along with standard deviation error bars. SVM Linear at 0.713 is the best one. Both GCN+Skip at 0.664 and Random Forest at 0.649 have lower accuracy than BiLSTM at 0.693 and Fusion at 0.671. More complex model has larger error bars.

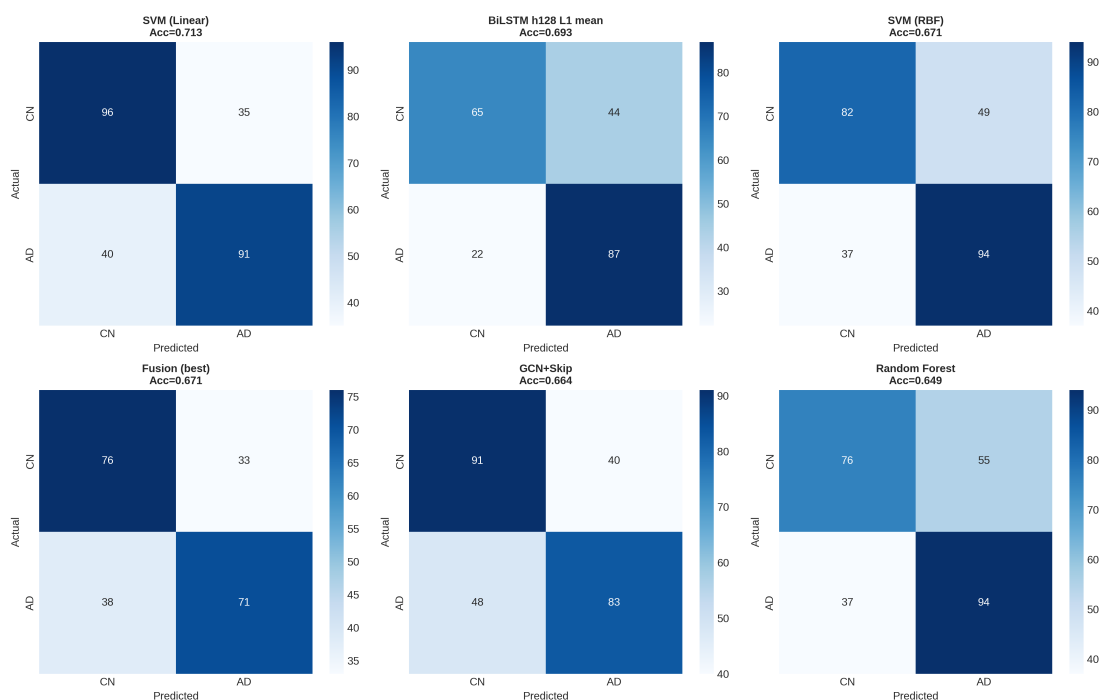


Figure B.13: Confusion Matrices for each of the Top 6 Models in a 2x3 Grid Format. From left to right, the models included in the top row include the following: SVM Linear (0.713), BiLSTM h128 (0.693), and SVM RBF (0.671). Fusion (0.671), GCN+Skip (0.664), and Random Forest (0.649).

B APPENDIX B: PER-MODEL DETAILED RESULTS

Random Forest (0.649) are on the bottom row. AD recall tends to be higher than CN recall in all models.

C Appendix C: Extended Explainability Analysis

This appendix includes expanded explainability analysis results: full ROI significance ranking, network-level importance distributions, cross-model importance correlation, connection importance matrix, and a four-panel summary graphic.

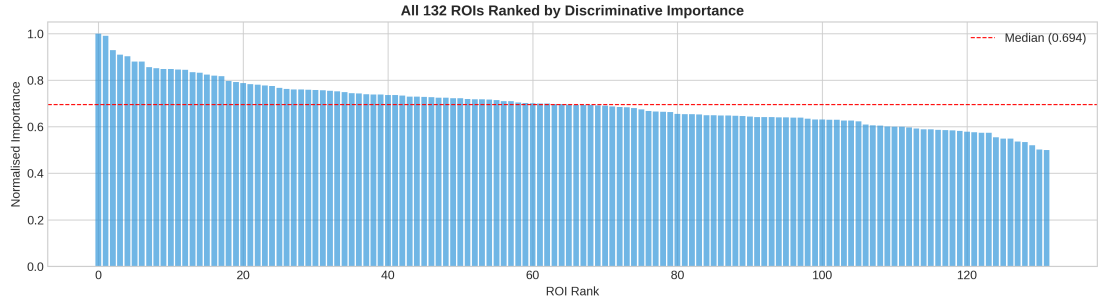


Figure C.1: SVM Linear coefficients averaged over five folds were used to rank all 132 ROIs based on normalized discriminative importance. The median importance (0.694) is shown by the red dashed line. There is no clear boundary between discriminative and non-discriminative areas, and importance steadily decreases from 1.0 to roughly 0.5.

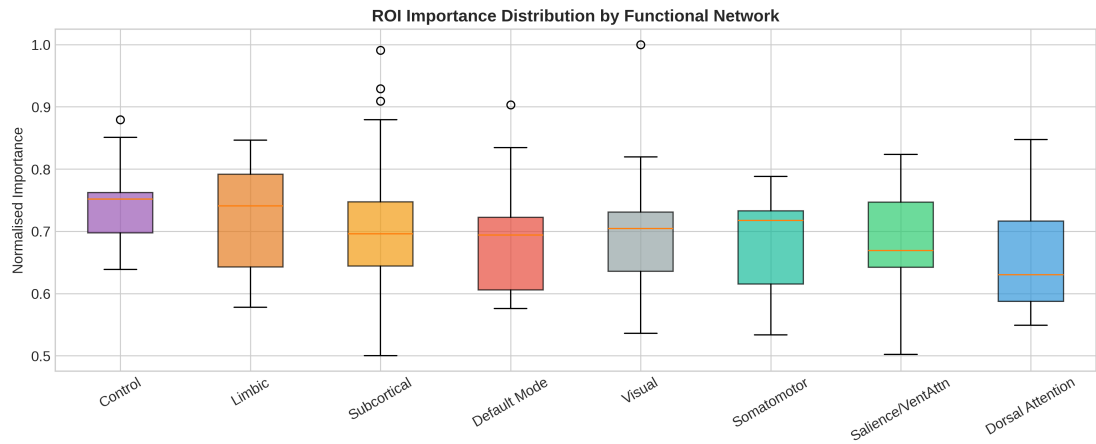


Figure C.2: The distribution of ROI relevance within each of the eight functional networks is displayed in a box plot. Although they have the fewest ROIs, control and limbic networks have the highest median significance. The largest spread is shown in subcortical and visual networks, where outliers have significance values greater than 0.9.

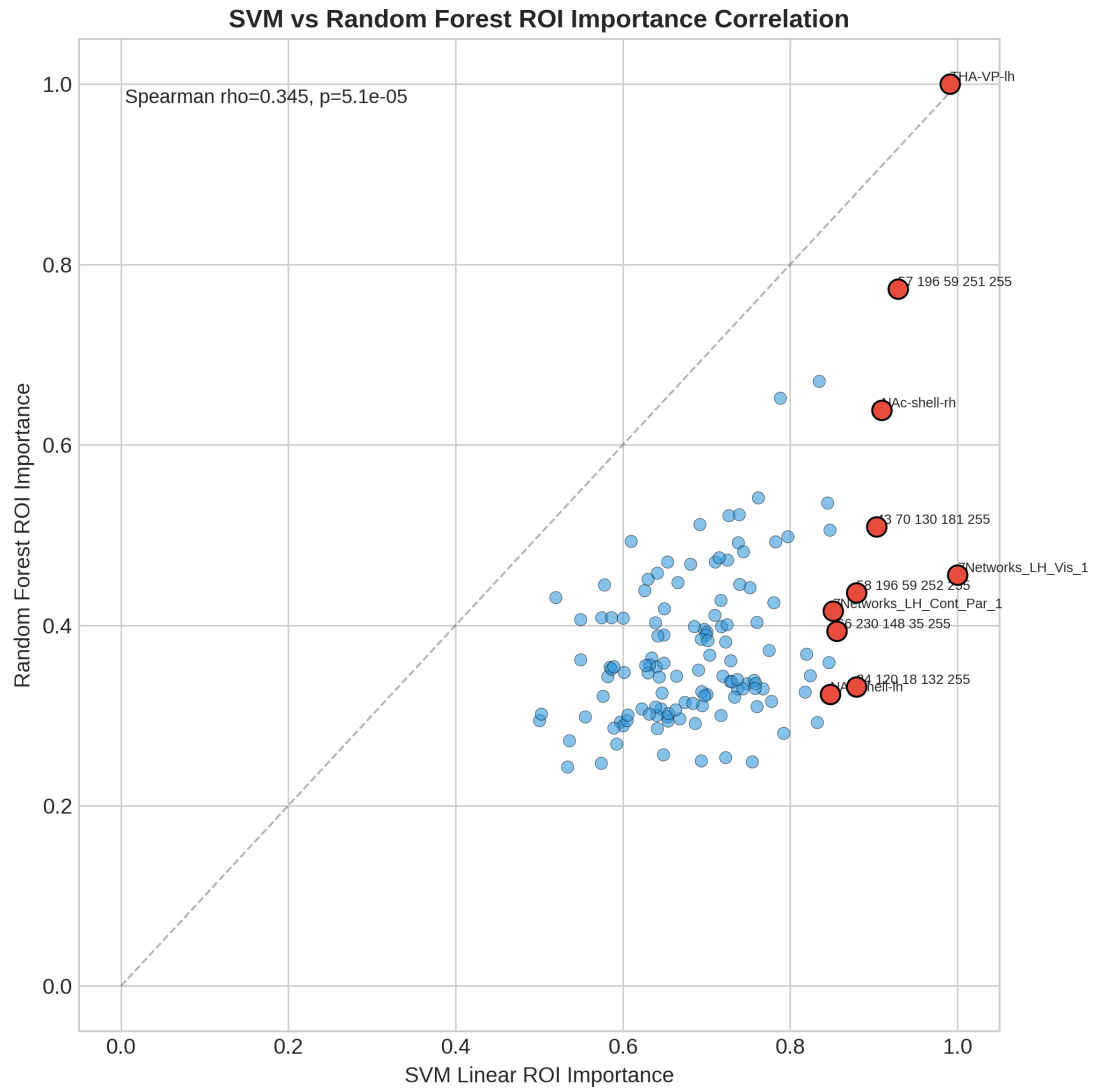


Figure C.3: A scatter plot comparing the normalized ROI relevance of SVM Linear (x-axis) and Random Forest (y-axis) for each of the 132 ROIs shows a statistically significant positive association with Spearman $\rho = 0.345$ ($p = 5.1e-05$). The top ten most discriminative ROIs by SVM significance are indicated by red circles. Perfect agreement is shown by the dashed diagonal line.

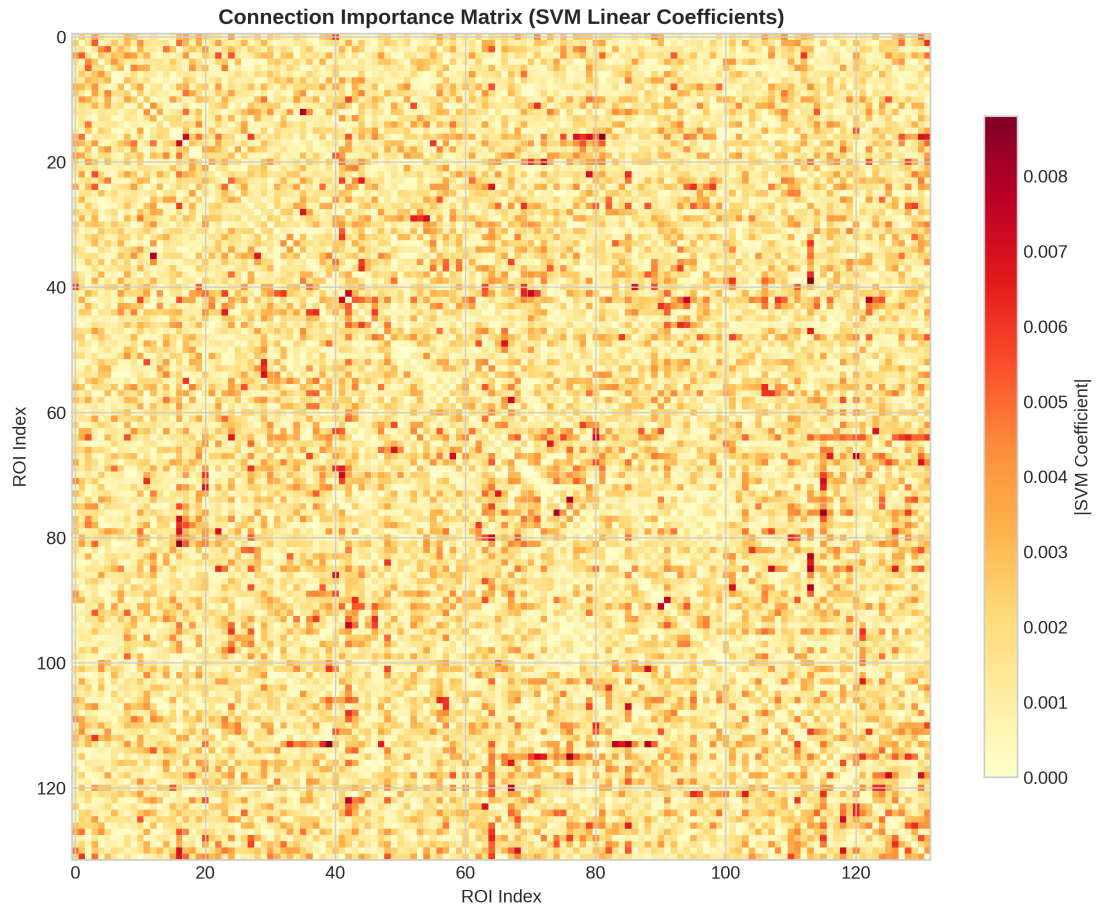


Figure C.4: Connection-level importance matrix (132 x 132) showing the absolute SVM Linear coefficient magnitude for every pairwise functional connection. Darker red cells indicate connections with higher discriminative importance. Visible block structure reflects the Schaefer network parcellation, with subcortical regions (indices 100-132) showing concentrated high-importance connections.

C APPENDIX C: EXTENDED EXPLAINABILITY ANALYSIS

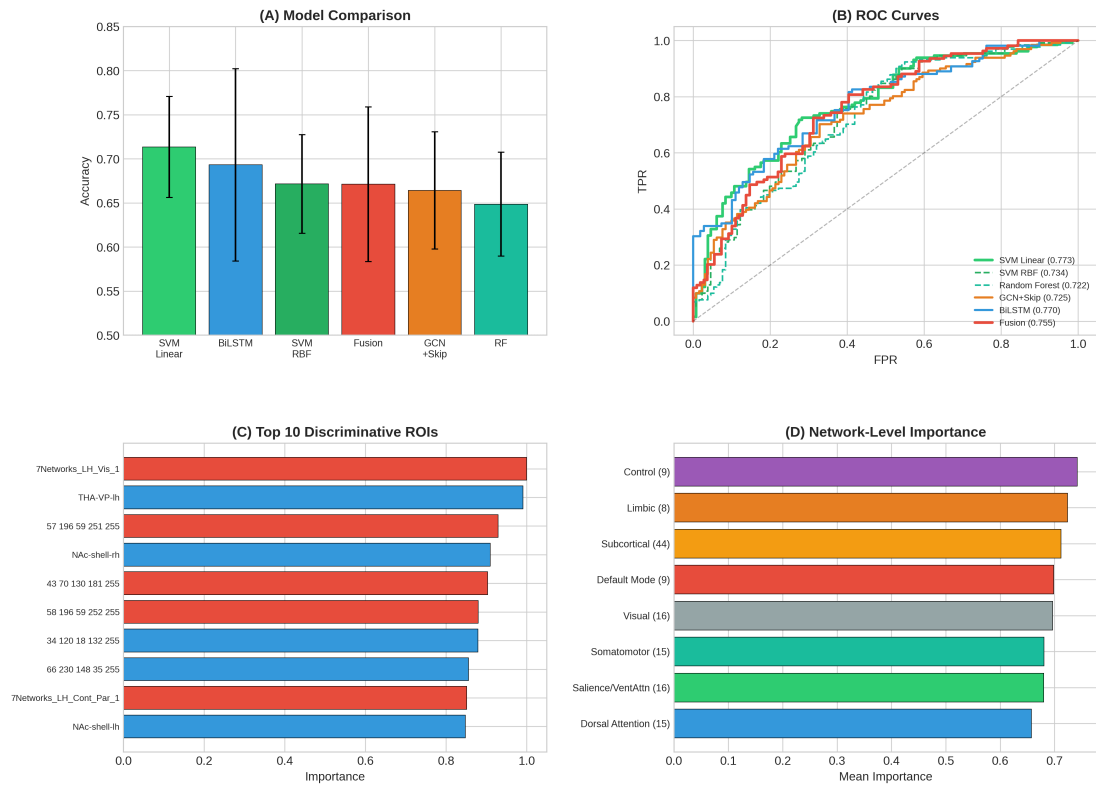


Figure C.5: Four-panel summary of key findings. (A) Classification accuracy comparison across all six best models with error bars. (B) ROC curves showing SVM Linear and BiLSTM nearly overlapping. (C) Top 10 most discriminative ROIs colour-coded by direction (red = higher in AD, blue = higher in CN). (D) Mean network-level importance with Control and Limbic networks ranking highest